# Custom printed OSM maps: Public transport and beyond

Dr Bartosz Fabianowski
bartosz@dobini.com

Dobini
www.dobini.com

Layout by orngjce223, CC-BY

# Printed stop information

- Ideal
  - Schedule
  - Strip map
  - Geographical map
- Reality
  - Poor presentation
  - Partial information
  - Outdated information
  - Expensive updates

Layout by orngjce223, CC-BY

# Poor schedules



GTT

FERMATA N. 380

ORARIO DEI PASSAGGI

Validità dal 01-07-2011

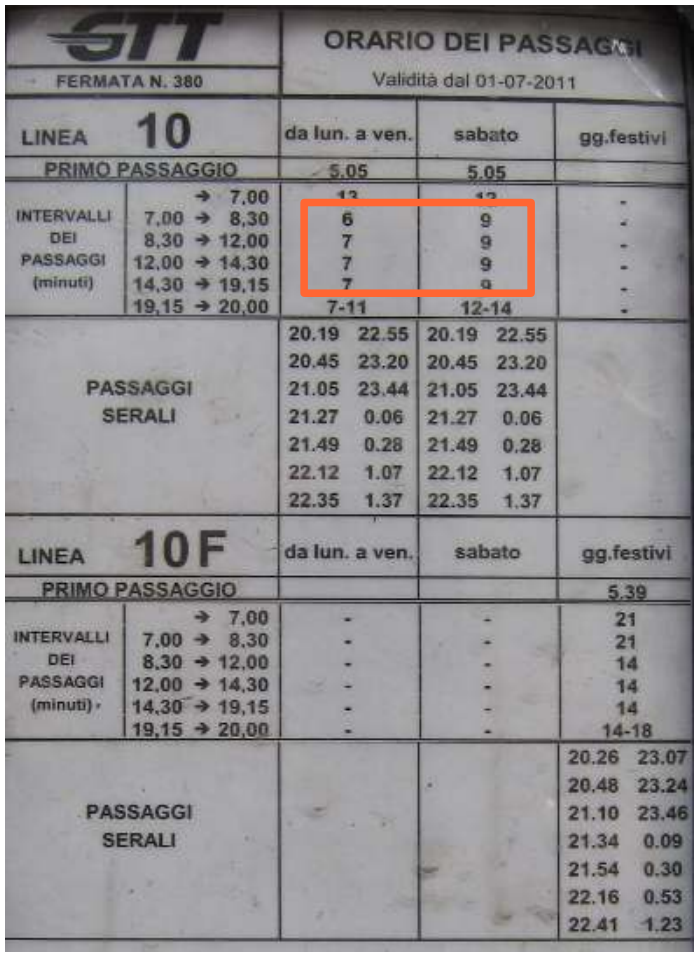
| LINEA 10 | | da lun. a ven. | sabato | gg.festivi |
|---|---|---|---|---|
| PRIMO PASSAGGIO | | 5.05 | 5.05 | |
| INTERVALLI DEI PASSAGGI (minuti) | → 7,00 | 13 | 12 | - |
| | 7,00 → 8,30 | 6 | 9 | - |
| | 8,30 → 12,00 | 7 | 9 | - |
| | 12,00 → 14,30 | 7 | 9 | - |
| | 14,30 → 19,15 | 7 | 9 | - |
| | 19,15 → 20,00 | 7-11 | 12-14 | - |
| PASSAGGI SERALI | | 20.19 22.55 | 20.19 22.55 | |
| | | 20.45 23.20 | 20.45 23.20 | |
| | | 21.05 23.44 | 21.05 23.44 | |
| | | 21.27 0.06 | 21.27 0.06 | |
| | | 21.49 0.28 | 21.49 0.28 | |
| | | 22.12 1.07 | 22.12 1.07 | |
| | | 22.35 1.37 | 22.35 1.37 | |

| LINEA 10F | | da lun. a ven. | sabato | gg.festivi |
|---|---|---|---|---|
| PRIMO PASSAGGIO | | | | 5.39 |
| INTERVALLI DEI PASSAGGI (minuti) | → 7,00 | - | - | 21 |
| | 7,00 → 8,30 | - | - | 21 |
| | 8,30 → 12,00 | - | - | 14 |
| | 12,00 → 14,30 | - | - | 14 |
| | 14,30 → 19,15 | - | - | 14 |
| | 19,15 → 20,00 | - | - | 14-18 |
| PASSAGGI SERALI | | | | 20.26 23.07 |
| | | | | 20.48 23.24 |
| | | | | 21.10 23.46 |
| | | | | 21.34 0.09 |
| | | | | 21.54 0.30 |
| | | | | 22.16 0.53 |
| | | | | 22.41 1.23 |

© Gruppo Torinese Trasporti S.p.A.

Layout by orngjce223, CC-BY

# Poor schedules



GTT

FERMATA N. 380

ORARIO DEI PASSAGGI

Validità dal 01-07-2011

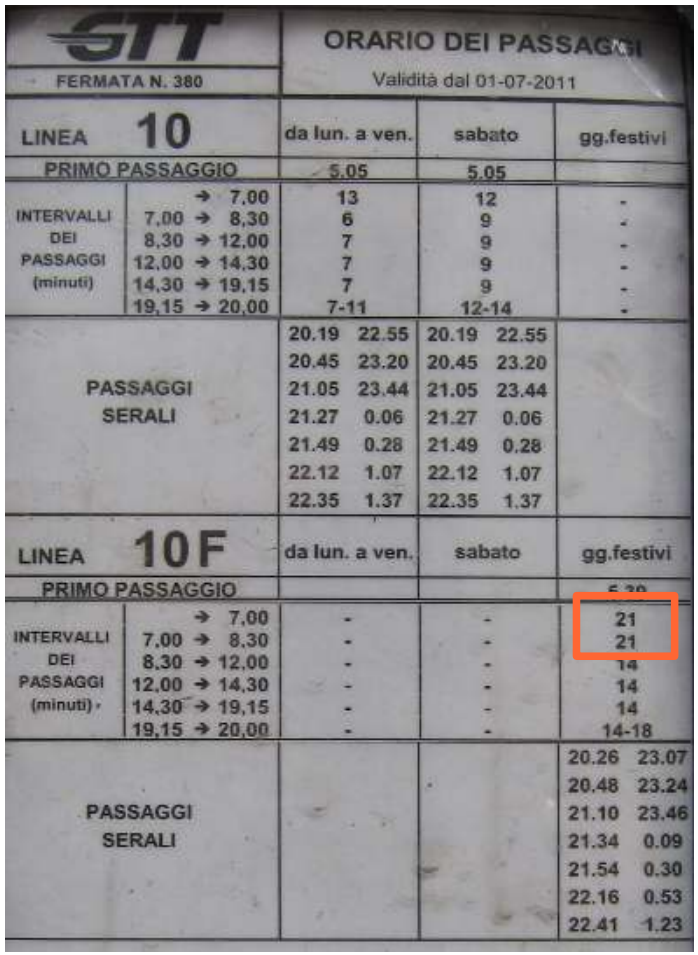
| LINEA 10 | | da lun. a ven. | sabato | gg.festivi |
|---|---|---|---|---|
| PRIMO PASSAGGIO | | 5.05 | 5.05 | |
| INTERVALLI DEI PASSAGGI (minuti) | → 7,00 | 13 | 13 | - |
| | 7,00 → 8,30 | 6 | 9 | - |
| | 8,30 → 12,00 | 7 | 9 | - |
| | 12,00 → 14,30 | 7 | 9 | - |
| | 14,30 → 19,15 | 7 | 9 | - |
| | 19,15 → 20,00 | 7-11 | 12-14 | - |
| PASSAGGI SERALI | | 20.19 22.55 | 20.19 22.55 | |
| | | 20.45 23.20 | 20.45 23.20 | |
| | | 21.05 23.44 | 21.05 23.44 | |
| | | 21.27 0.06 | 21.27 0.06 | |
| | | 21.49 0.28 | 21.49 0.28 | |
| | | 22.12 1.07 | 22.12 1.07 | |
| | | 22.35 1.37 | 22.35 1.37 | |

| LINEA 10F | | da lun. a ven. | sabato | gg.festivi |
|---|---|---|---|---|
| PRIMO PASSAGGIO | | | | 5.39 |
| INTERVALLI DEI PASSAGGI (minuti) | → 7,00 | - | - | 21 |
| | 7,00 → 8,30 | - | - | 21 |
| | 8,30 → 12,00 | - | - | 14 |
| | 12,00 → 14,30 | - | - | 14 |
| | 14,30 → 19,15 | - | - | 14 |
| | 19,15 → 20,00 | - | - | 14-18 |
| PASSAGGI SERALI | | | | 20.26 23.07 |
| | | | | 20.48 23.24 |
| | | | | 21.10 23.46 |
| | | | | 21.34 0.09 |
| | | | | 21.54 0.30 |
| | | | | 22.16 0.53 |
| | | | | 22.41 1.23 |

© Gruppo Torinese Trasporti S.p.A.

Layout by orngjce223, CC-BY

# Poor schedules



GTT

FERMATA N. 380

ORARIO DEI PASSAGGI

Validità dal 01-07-2011

| LINEA 10 | | da lun. a ven. | sabato | gg.festivi |
|---|---|---|---|---|
| PRIMO PASSAGGIO | | 5.05 | 5.05 | |
| INTERVALLI DEI PASSAGGI (minuti) | → 7,00 | 13 | 12 | - |
| | 7,00 → 8,30 | 6 | 9 | - |
| | 8,30 → 12,00 | 7 | 9 | - |
| | 12,00 → 14,30 | 7 | 9 | - |
| | 14,30 → 19,15 | 7 | 9 | - |
| | 19,15 → 20,00 | 7-11 | 12-14 | - |
| PASSAGGI SERALI | | 20.19 22.55 | 20.19 22.55 | |
| | | 20.45 23.20 | 20.45 23.20 | |
| | | 21.05 23.44 | 21.05 23.44 | |
| | | 21.27 0.06 | 21.27 0.06 | |
| | | 21.49 0.28 | 21.49 0.28 | |
| | | 22.12 1.07 | 22.12 1.07 | |
| | | 22.35 1.37 | 22.35 1.37 | |

| LINEA 10F | | da lun. a ven. | sabato | gg.festivi |
|---|---|---|---|---|
| PRIMO PASSAGGIO | | | | |
| INTERVALLI DEI PASSAGGI (minuti) | → 7,00 | - | - | 21 |
| | 7,00 → 8,30 | - | - | 21 |
| | 8,30 → 12,00 | - | - | 14 |
| | 12,00 → 14,30 | - | - | 14 |
| | 14,30 → 19,15 | - | - | 14 |
| | 19,15 → 20,00 | - | - | 14-18 |
| PASSAGGI SERALI | | | | 20.26 23.07 |
| | | | | 20.48 23.24 |
| | | | | 21.10 23.46 |
| | | | | 21.34 0.09 |
| | | | | 21.54 0.30 |
| | | | | 22.16 0.53 |
| | | | | 22.41 1.23 |

© Gruppo Torinese Trasporti S.p.A.

Layout by orngjce223, CC-BY

# Poor schedules



94

## P.TA VOLTA - CADORNA FN

### Fermata Colonne di S. Lorenzo

Orario in vigore dal 06/09/2010

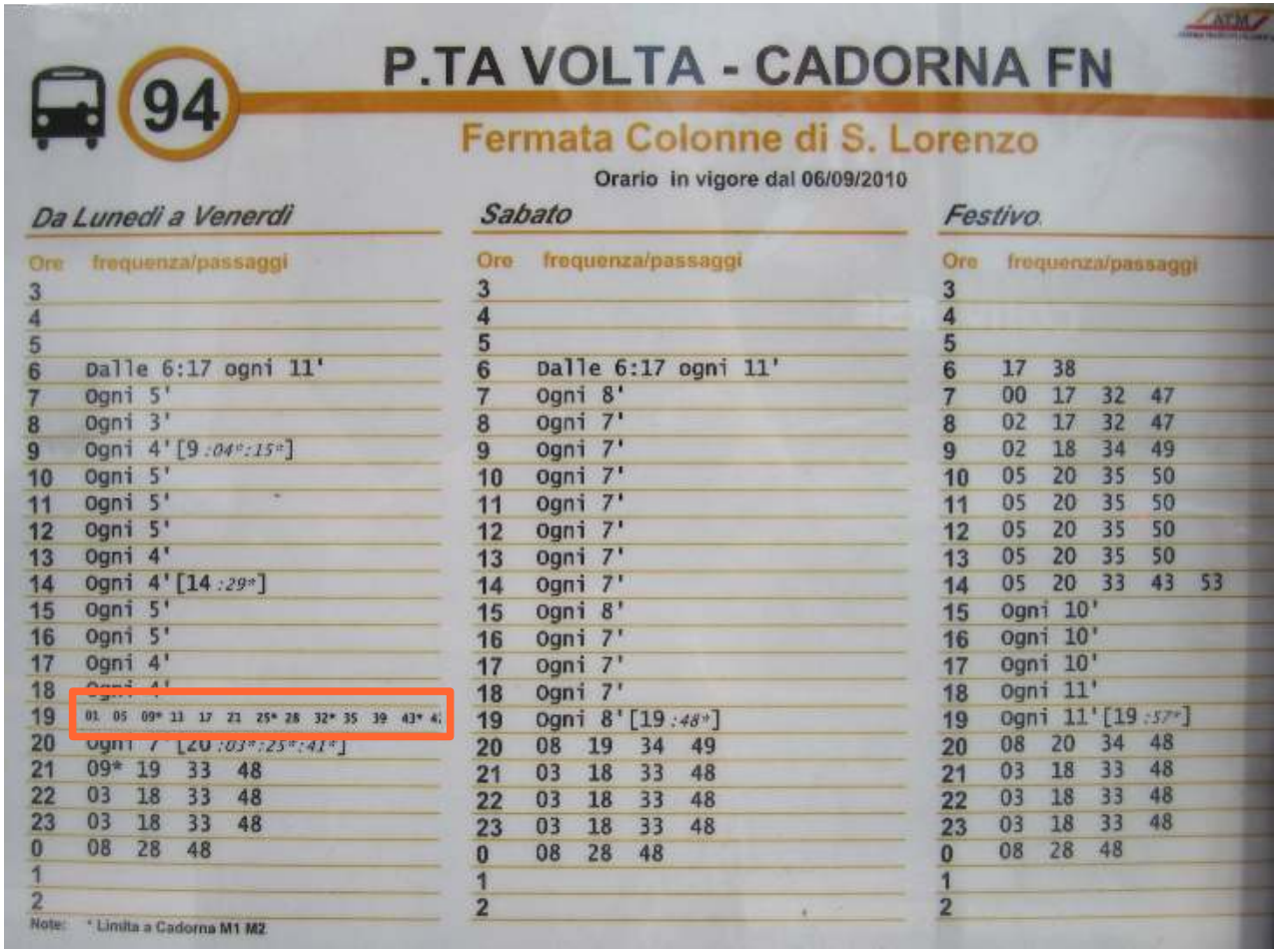
| Da Lunedi a Venerdi | | Sabato | | Festivo | |
|---|---|---|---|---|---|
| Ore | frequenza/passaggi | Ore | frequenza/passaggi | Ore | frequenza/passaggi |
| 3 | | 3 | | 3 | |
| 4 | | 4 | | 4 | |
| 5 | | 5 | | 5 | |
| 6 | Dalle 6:17 ogni 11' | 6 | Dalle 6:17 ogni 11' | 6 | 17 38 |
| 7 | Ogni 5' | 7 | Ogni 8' | 7 | 00 17 32 47 |
| 8 | Ogni 3' | 8 | Ogni 7' | 8 | 02 17 32 47 |
| 9 | Ogni 4' [9 :04*:15*] | 9 | Ogni 7' | 9 | 02 18 34 49 |
| 10 | Ogni 5' | 10 | Ogni 7' | 10 | 05 20 35 50 |
| 11 | Ogni 5' | 11 | Ogni 7' | 11 | 05 20 35 50 |
| 12 | Ogni 5' | 12 | Ogni 7' | 12 | 05 20 35 50 |
| 13 | Ogni 4' | 13 | Ogni 7' | 13 | 05 20 35 50 |
| 14 | Ogni 4' [14 :29*] | 14 | Ogni 7' | 14 | 05 20 33 43 53 |
| 15 | Ogni 5' | 15 | Ogni 8' | 15 | Ogni 10' |
| 16 | Ogni 5' | 16 | Ogni 7' | 16 | Ogni 10' |
| 17 | Ogni 4' | 17 | Ogni 7' | 17 | Ogni 10' |
| 18 | Ogni 4' | 18 | Ogni 7' | 18 | Ogni 11' |
| 19 | 01 05 09* 11 17 21 25* 28 32* 35 39 43* 4 | 19 | Ogni 8' [19 :48*] | 19 | Ogni 11' [19 :57*] |
| 20 | Ogni 7' [20 :03*:25*:41*] | 20 | 08 19 34 49 | 20 | 08 20 34 48 |
| 21 | 09* 19 33 48 | 21 | 03 18 33 48 | 21 | 03 18 33 48 |
| 22 | 03 18 33 48 | 22 | 03 18 33 48 | 22 | 03 18 33 48 |
| 23 | 03 18 33 48 | 23 | 03 18 33 48 | 23 | 03 18 33 48 |
| 0 | 08 28 48 | 0 | 08 28 48 | 0 | 08 28 48 |
| 1 | | 1 | | 1 | |
| 2 | | 2 | | 2 | |

Note: * Limita a Cadorna M1 M2

© Azienda Trasporti Milanesi S.p.A.

Layout by orngjce223, CC-BY

# Poor schedules



94

## P.TA VOLTA - CADORNA FN

### Fermata Colonne di S. Lorenzo

Orario in vigore dal 06/09/2010

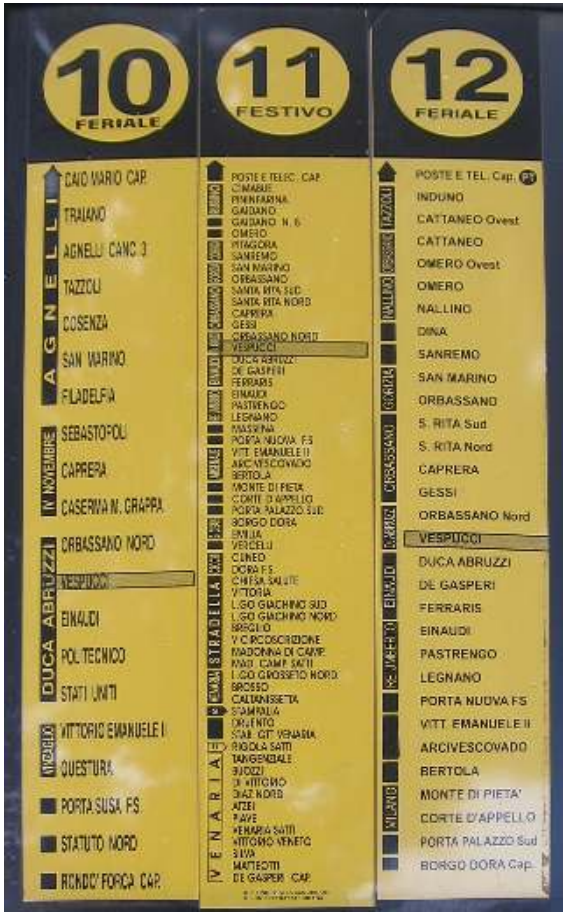
| Da Lunedi a Venerdi | | Sabato | | Festivo | |
|---|---|---|---|---|---|
| Ore | frequenza/passaggi | Ore | frequenza/passaggi | Ore | frequenza/passaggi |
| 3 | | 3 | | 3 | |
| 4 | | 4 | | 4 | |
| 5 | | 5 | | 5 | |
| 6 | Dalle 6:17 ogni 11' | 6 | Dalle 6:17 ogni 11' | 6 | 17 38 |
| 7 | Ogni 5' | 7 | Ogni 8' | 7 | 00 17 32 47 |
| 8 | Ogni 3' | 8 | Ogni 7' | 8 | 02 17 32 47 |
| 9 | Ogni 4' [9 :04* :15*] | 9 | Ogni 7' | 9 | 02 18 34 49 |
| 10 | Ogni 5' | 10 | Ogni 7' | 10 | 05 20 35 50 |
| 11 | Ogni 5' | 11 | Ogni 7' | 11 | 05 20 35 50 |
| 12 | Ogni 5' | 12 | Ogni 7' | 12 | 05 20 35 50 |
| 13 | Ogni 4' | 13 | Ogni 7' | 13 | 05 20 35 50 |
| 14 | Ogni 4' [14 :29*] | 14 | Ogni 7' | 14 | 05 20 33 43 53 |
| 15 | Ogni 5' | 15 | Ogni 8' | 15 | Ogni 10' |
| 16 | Ogni 5' | 16 | Ogni 7' | 16 | Ogni 10' |
| 17 | Ogni 4' | 17 | Ogni 7' | 17 | Ogni 10' |
| 18 | [illegible] | 18 | Ogni 7' | 18 | Ogni 11' |
| 19 | 01 05 09* 11 17 21 25* 28 32* 35 39 43* 4[illegible] | 19 | Ogni 8' [19 :48*] | 19 | Ogni 11' [19 :57*] |
| 20 | Ogni 7' [20 :03* :25* :41*] | 20 | 08 19 34 49 | 20 | 08 20 34 48 |
| 21 | 09* 19 33 48 | 21 | 03 18 33 48 | 21 | 03 18 33 48 |
| 22 | 03 18 33 48 | 22 | 03 18 33 48 | 22 | 03 18 33 48 |
| 23 | 03 18 33 48 | 23 | 03 18 33 48 | 23 | 03 18 33 48 |
| 0 | 08 28 48 | 0 | 08 28 48 | 0 | 08 28 48 |
| 1 | | 1 | | 1 | |
| 2 | | 2 | | 2 | |

Note: * Limita a Cadorna M1 M2

© Azienda Trasporti Milanesi S.p.A.

Layout by orngjce223, CC-BY

# Poor strip maps

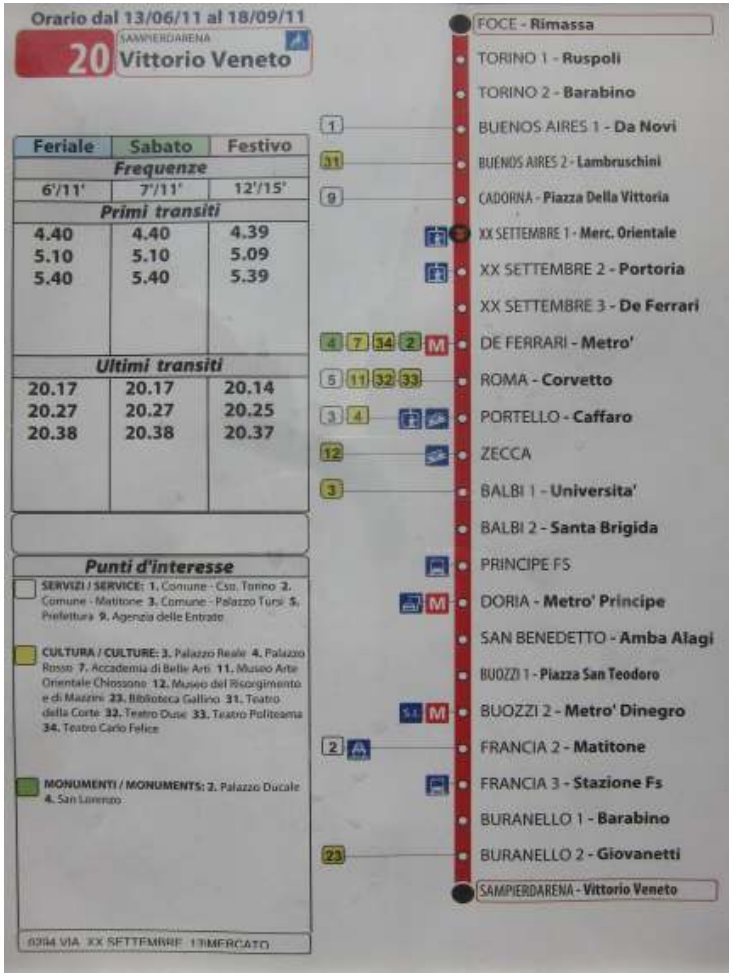



© Gruppo Torinese Trasporti S.p.A.



Layout by orngjce223, CC-BY

# Poor strip maps

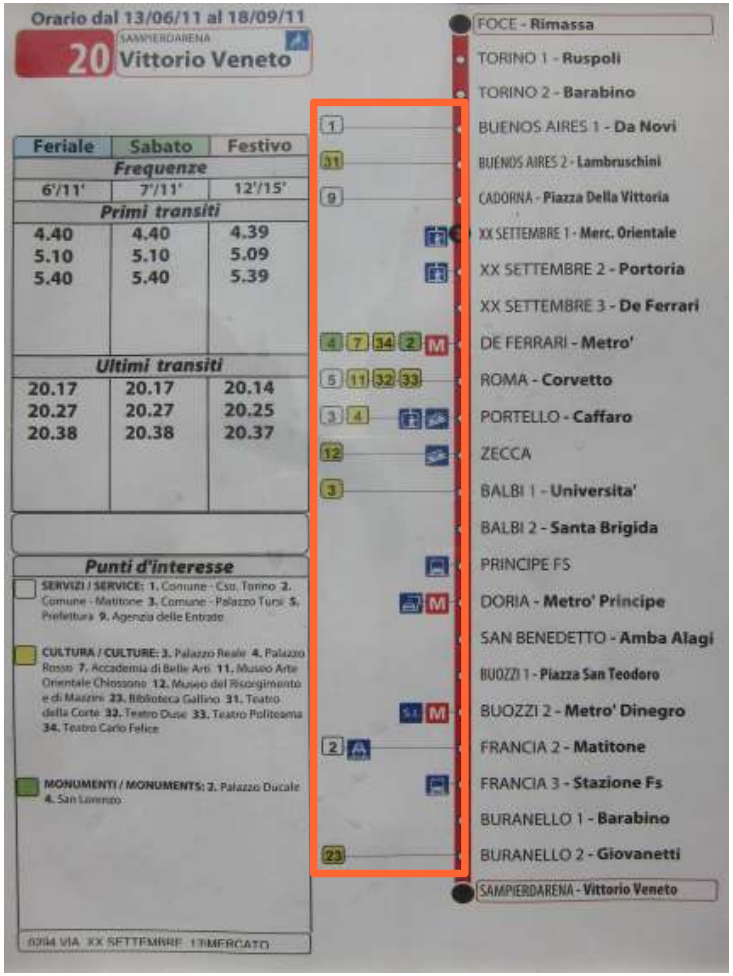



© Azienda Mobilità e Trasporti S.p.A.



Layout by orngjce223, CC-BY

# Poor strip maps

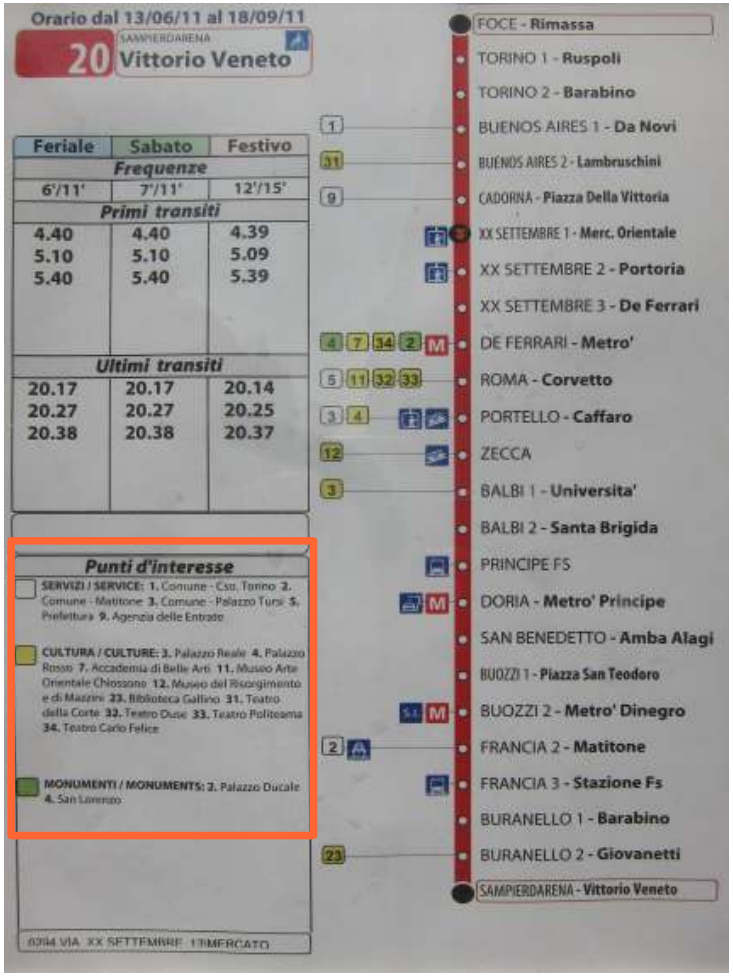



© Azienda Mobilità e Trasporti S.p.A.



Layout by orngjce223, CC-BY

# Poor strip maps

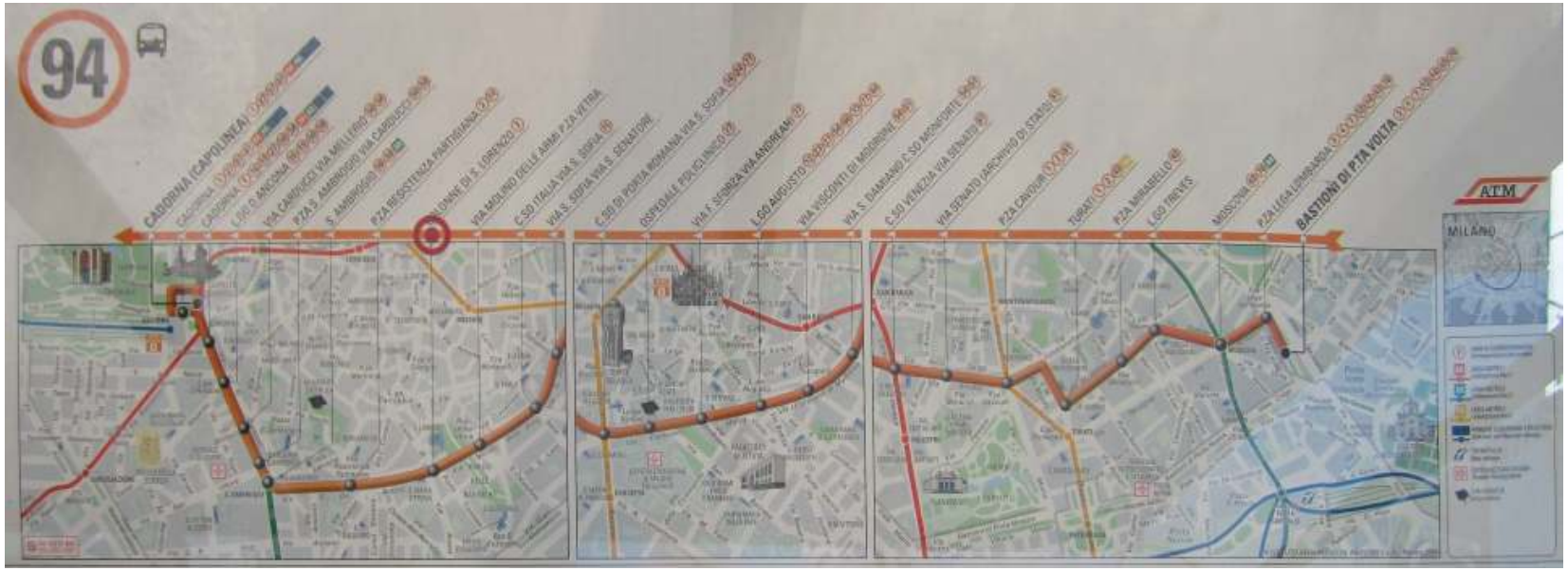



© Azienda Mobilità e Trasporti S.p.A.



Layout by orngjce223, CC-BY

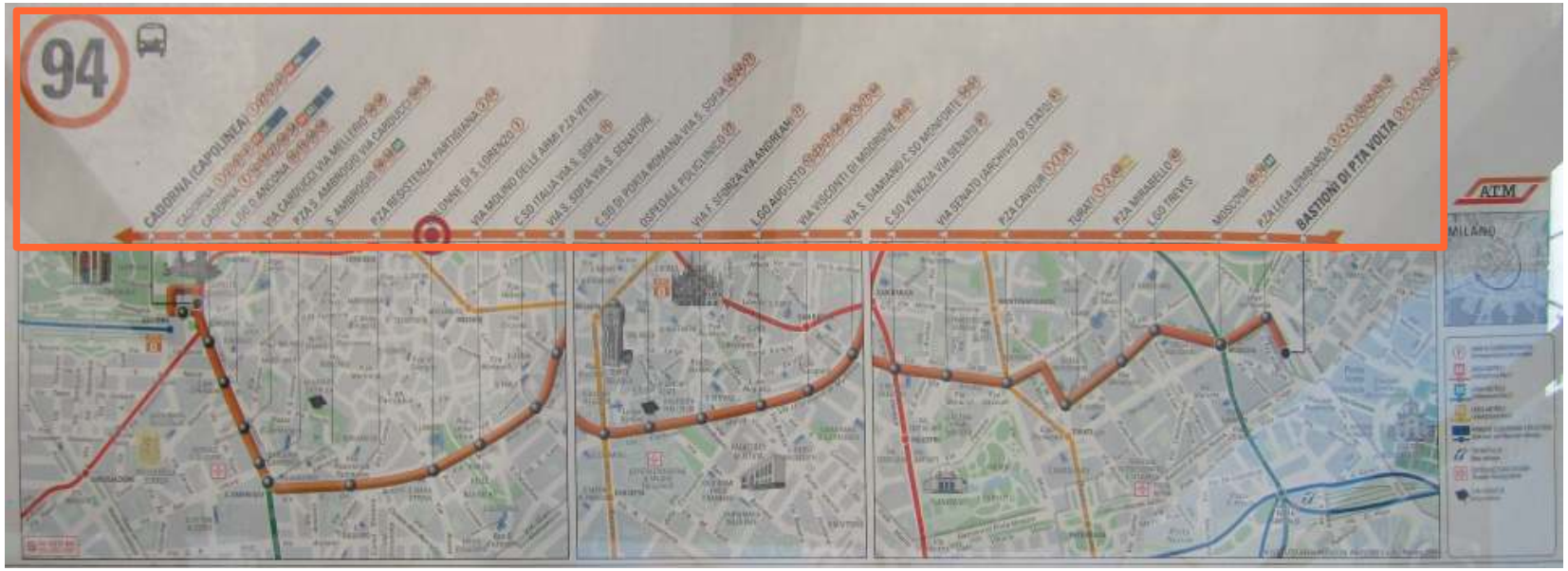
# Poor maps




© Azienda Trasporti Milanesi S.p.A.

Layout by orngjce223, CC-BY

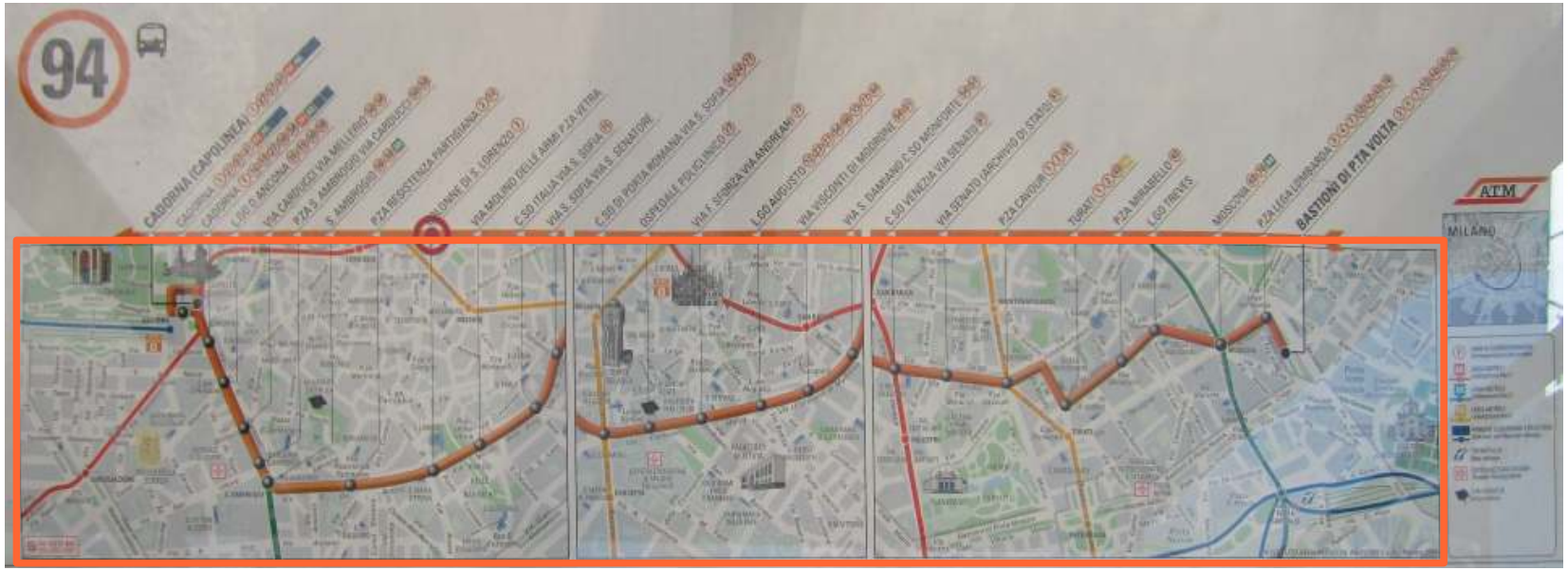
# Poor maps




© Azienda Trasporti Milanesi S.p.A.

Layout by orngjce223, CC-BY

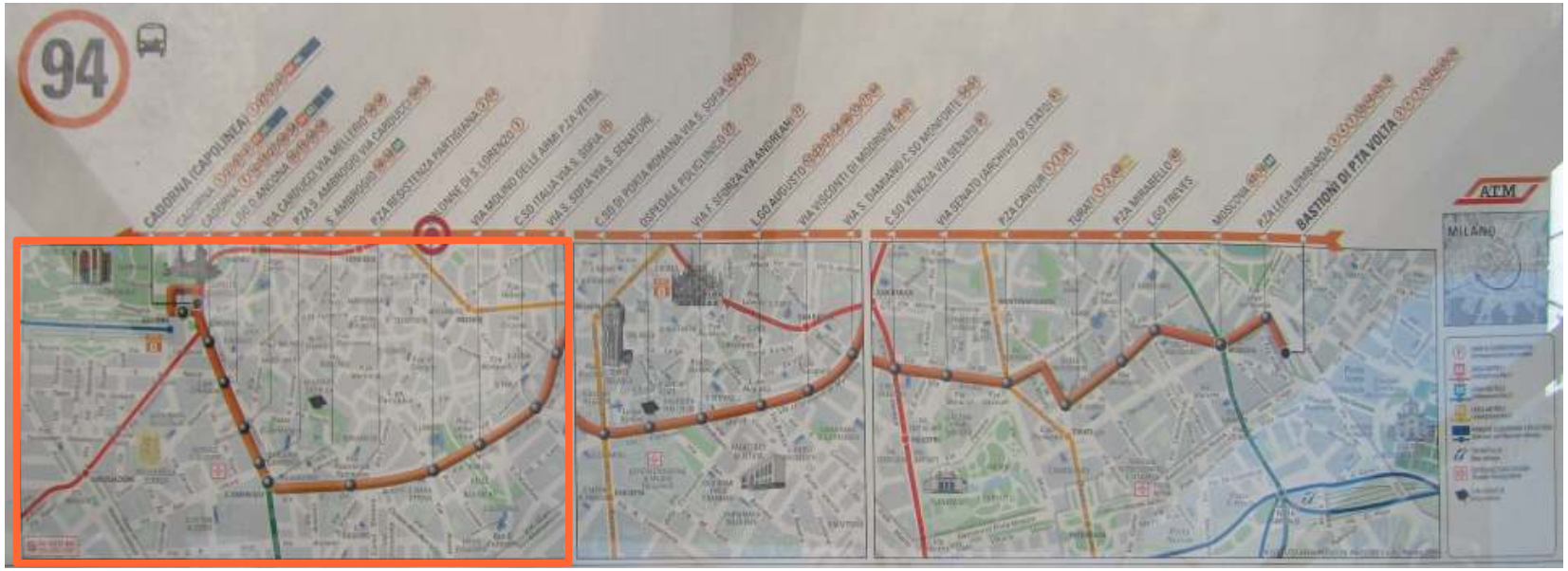
# Poor maps




© Azienda Trasporti Milanesi S.p.A.

Layout by orngjce223, CC-BY

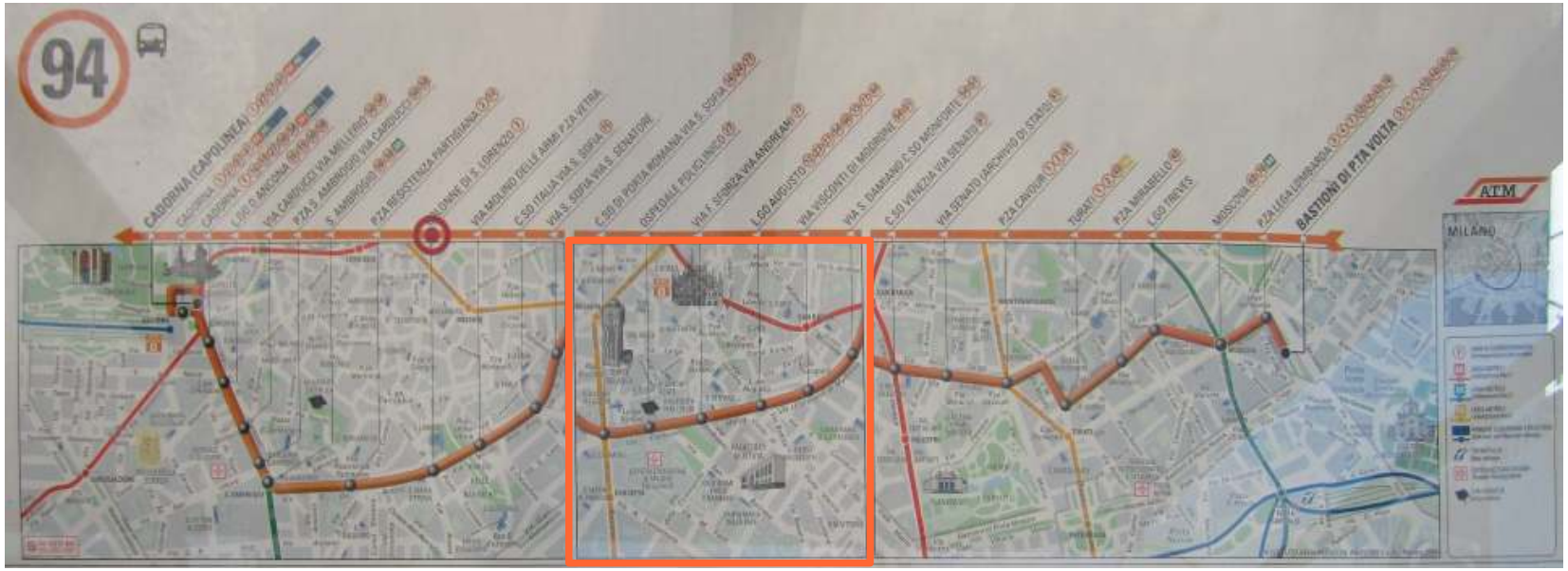
# Poor maps




© Azienda Trasporti Milanesi S.p.A.

Layout by orngjce223, CC-BY

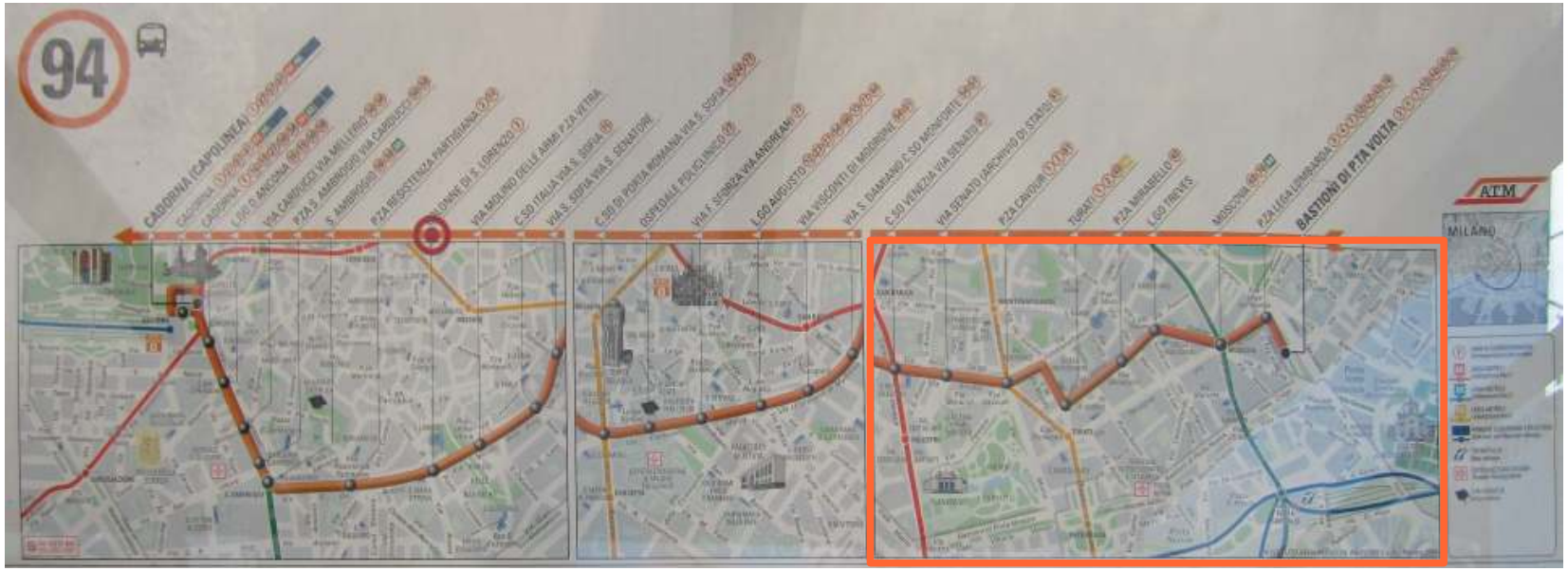
# Poor maps




© Azienda Trasporti Milanesi S.p.A.

Layout by orngjce223, CC-BY

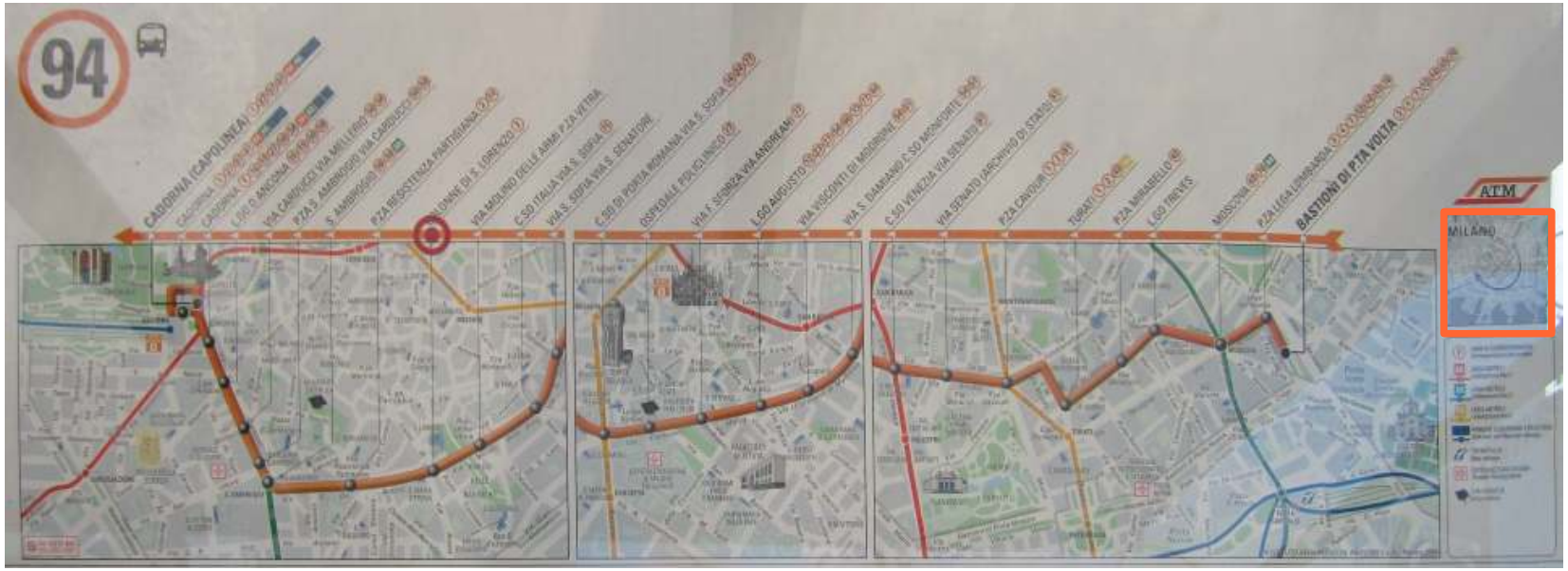
# Poor maps




© Azienda Trasporti Milanesi S.p.A.

Layout by orngjce223, CC-BY

# Poor maps




© Azienda Trasporti Milanesi S.p.A.

Layout by orngjce223, CC-BY

# Expensive to maintain maps



© Gruppo Torinese Trasporti S.p.A.



Layout by orngjce223, CC-BY

# Information sources

- Transport operator
  - Stops and routes → Operational details
  - Transport network context
  - Infrequent updates
- OSM
  - Stops and routes → Geographical details
  - Geographical context
  - Continuous updates

Layout by orngjce223, CC-BY

# Information presentation

- Printed stop placards
  - Schedule
  - Strip map
  - Overview map
  - Close-up map
- Using operator and OSM data
  - Driven by changes to operator data
  - Incorporating changes to OSM data

Layout by orngjce223, CC-BY

# Placard generation

- Automatic
  - No manual map layout
  - No manual selection of map regions
  - Fast and inexpensive updates
  - No manual ability to fix problems
- Components combining operator and OSM data
  - Strip map
  - Overview and close-up maps

Layout by orngjce223, CC-BY

# Operator and OSM data

- Transport operator
  - Schedules
  - Connections
  - Spelling and naming inconsistencies
  - Operator, route reference, stop reference
- OSM
  - Ways along route
  - Stop locations and attributes
  - Operator, route reference, stop reference

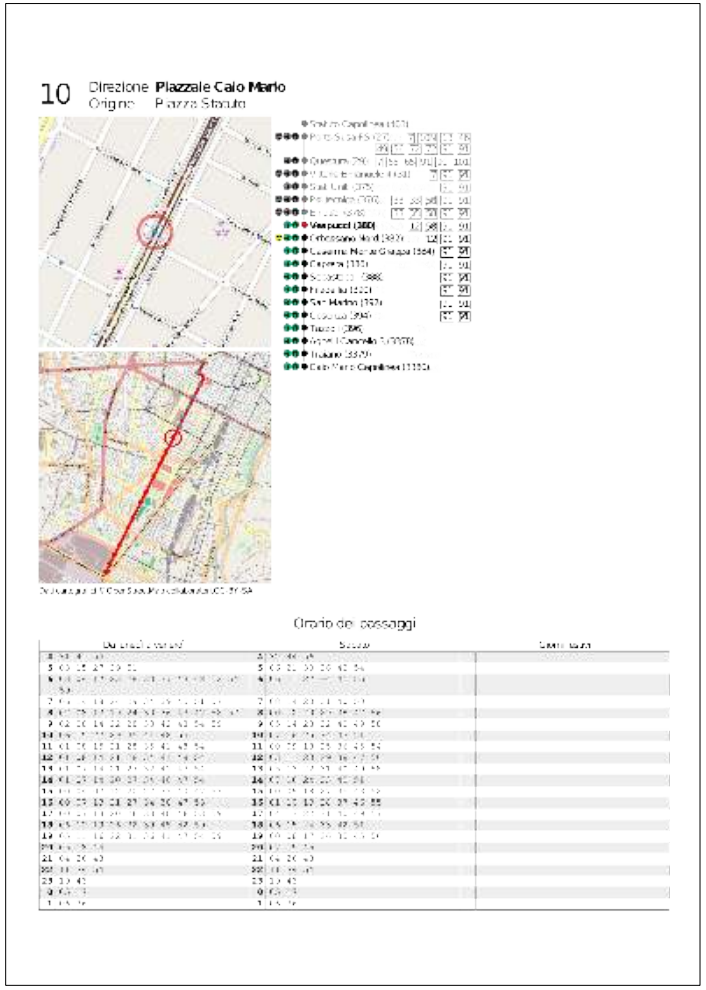
Layout by orngjce223, CC-BY

# Operator and OSM data

- Transport operator
  - Schedules
  - Connections
  - Spelling and naming inconsistencies
  - Operator, route reference, stop reference
- OSM
  - Ways along route
  - Stop locations and attributes
  - Operator, route reference, stop reference

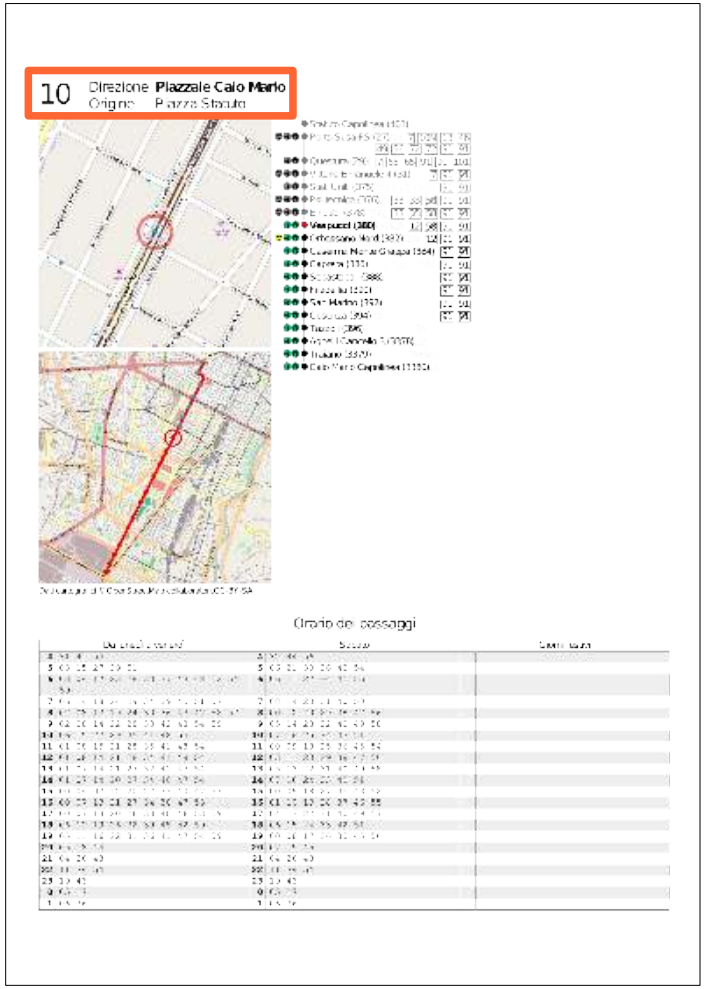
Layout by orngjce223, CC-BY

# Printed stop placards




Layout by orngjce223, CC-BY

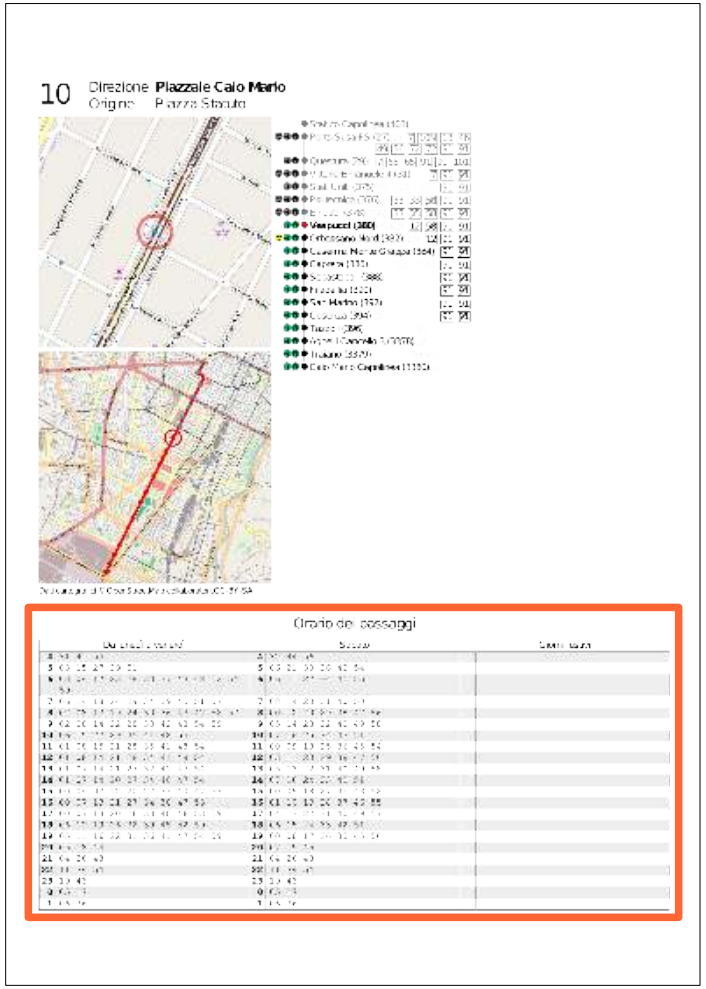
# Printed stop placards






Layout by orngjce223, CC-BY

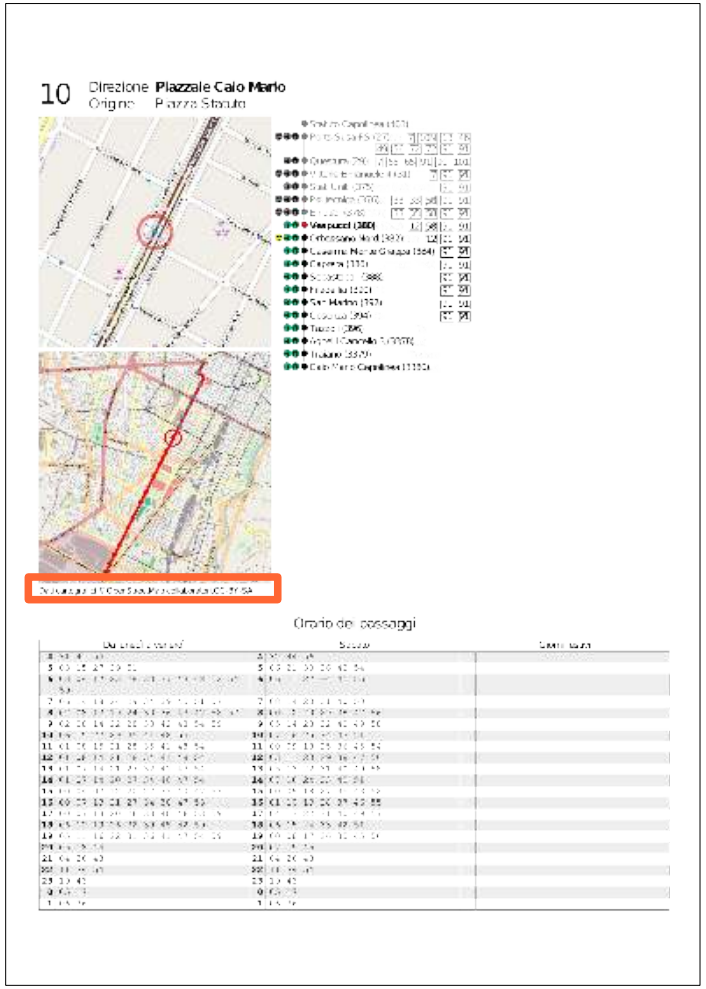
# Printed stop placards





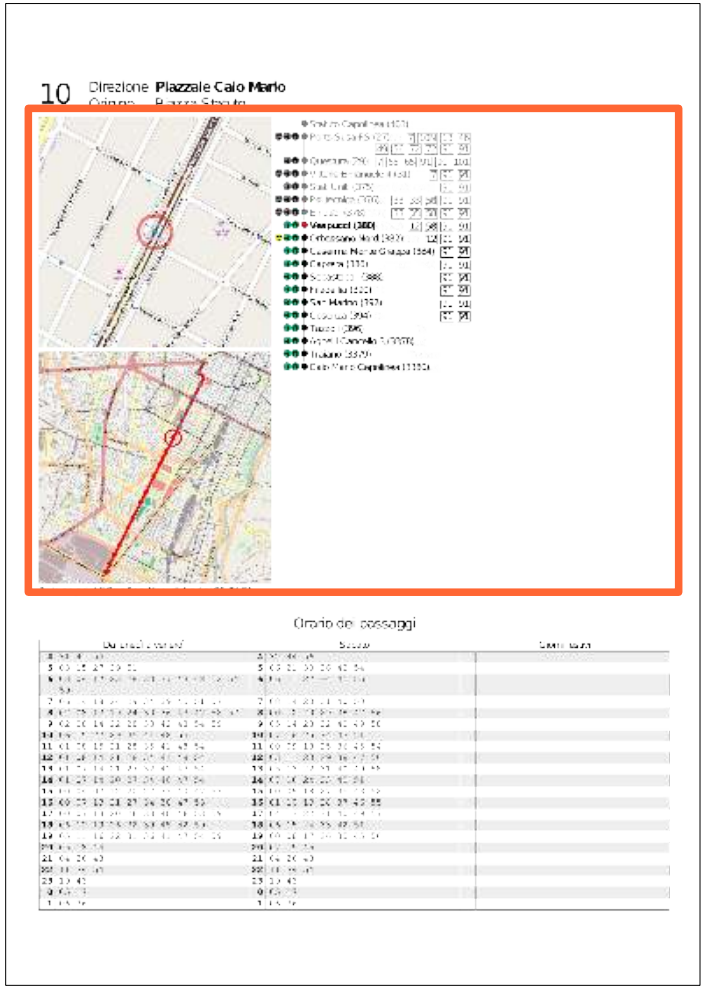

Layout by orngjce223, CC-BY

# Printed stop placards



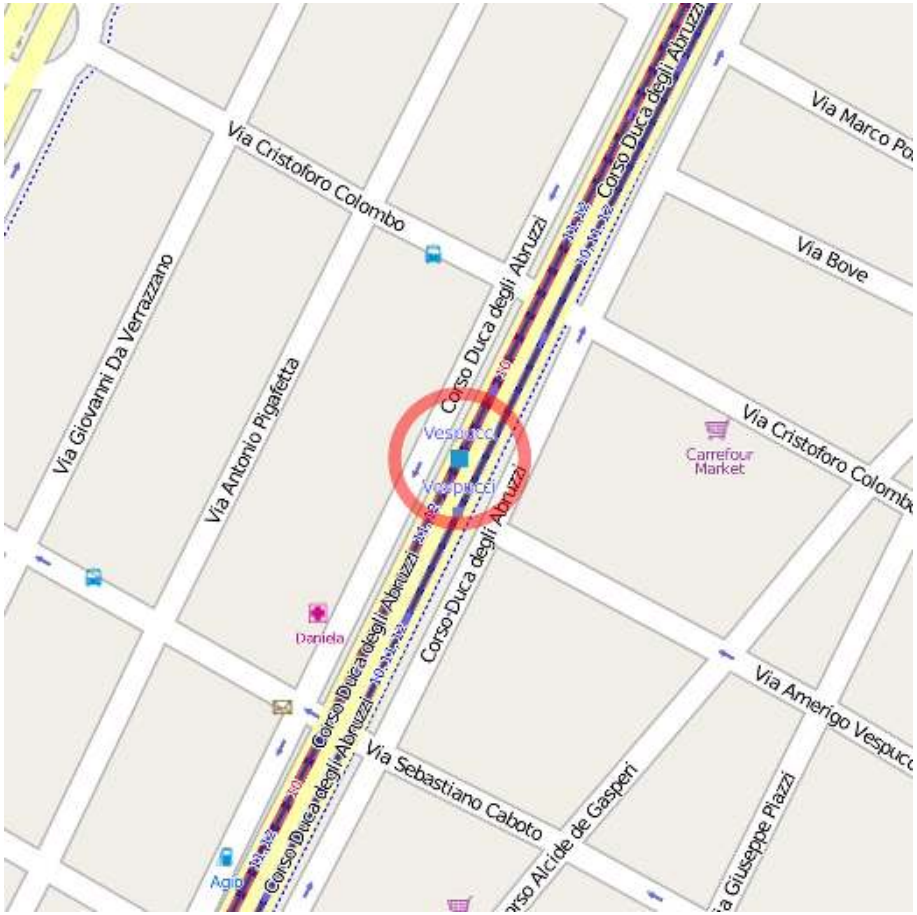

Layout by orngjce223, CC-BY

# Printed stop placards



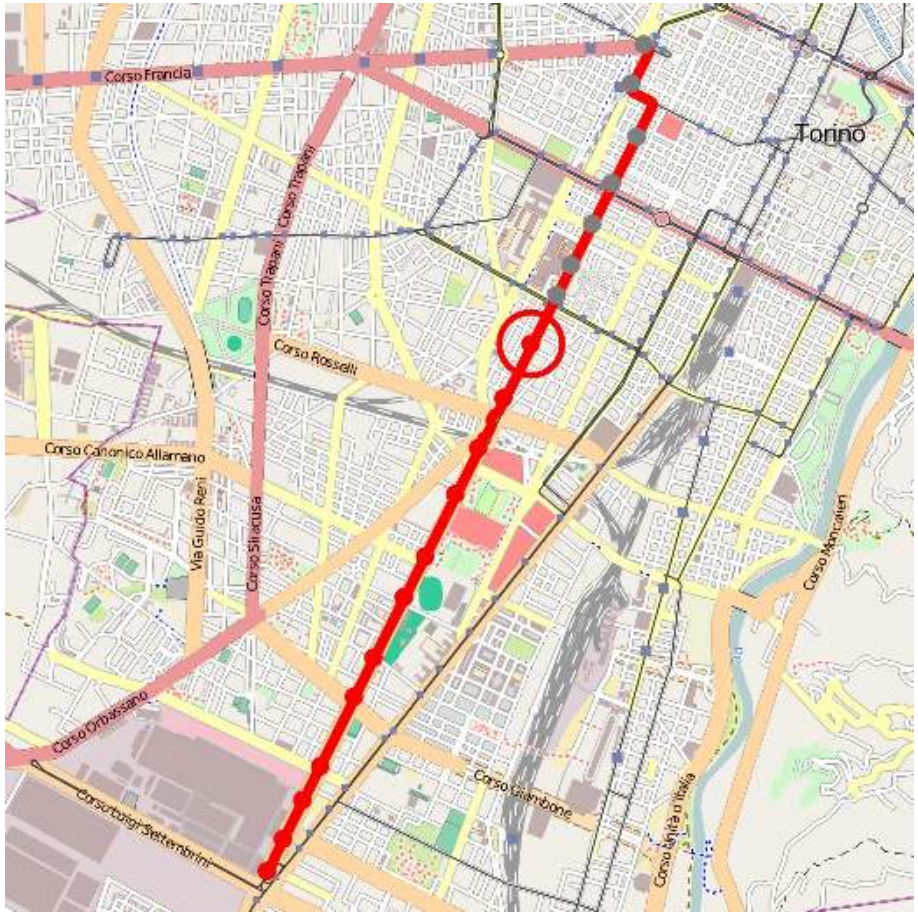

Layout by orngjce223, CC-BY

# Close-up map



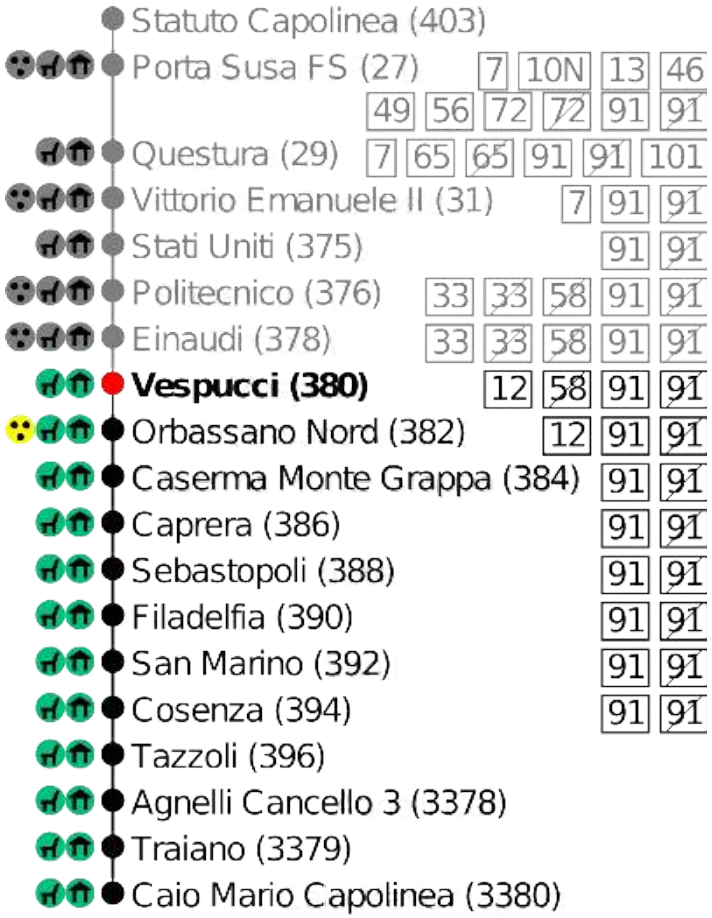

Layout by orngjce223, CC-BY

# Overview map



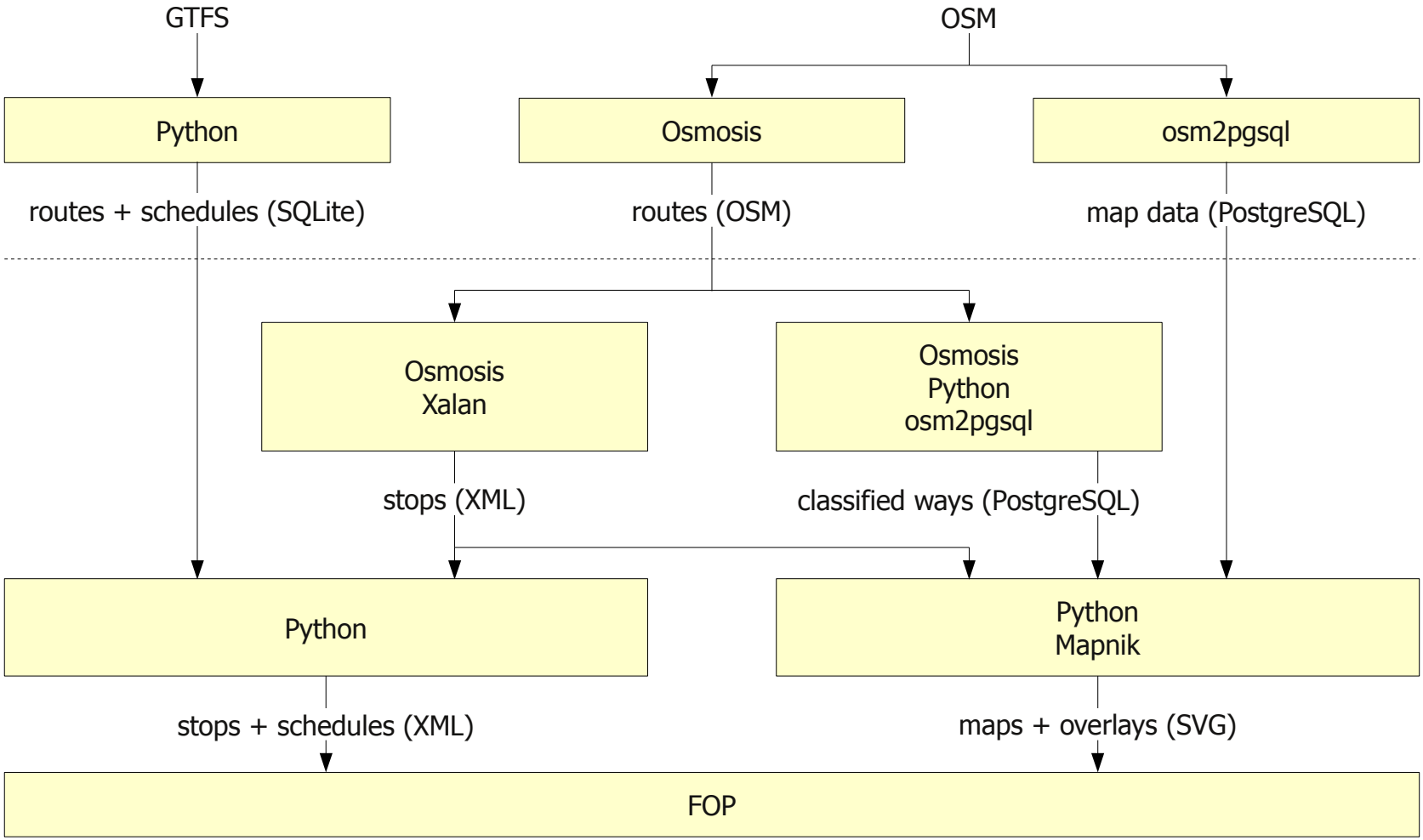

Layout by orngjce223, CC-BY

# Strip map



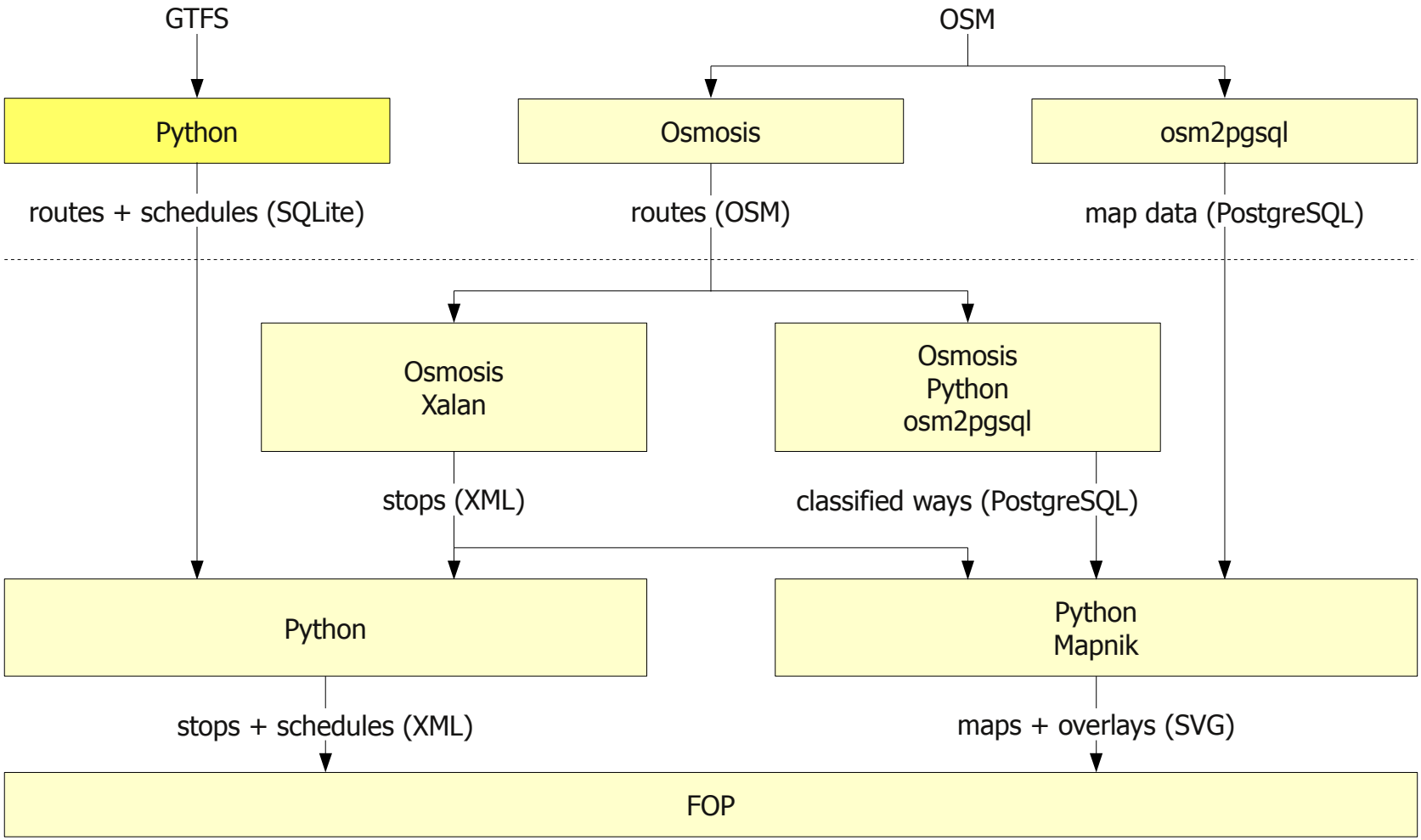

Layout by orngjce223, CC-BY

# Toolchain



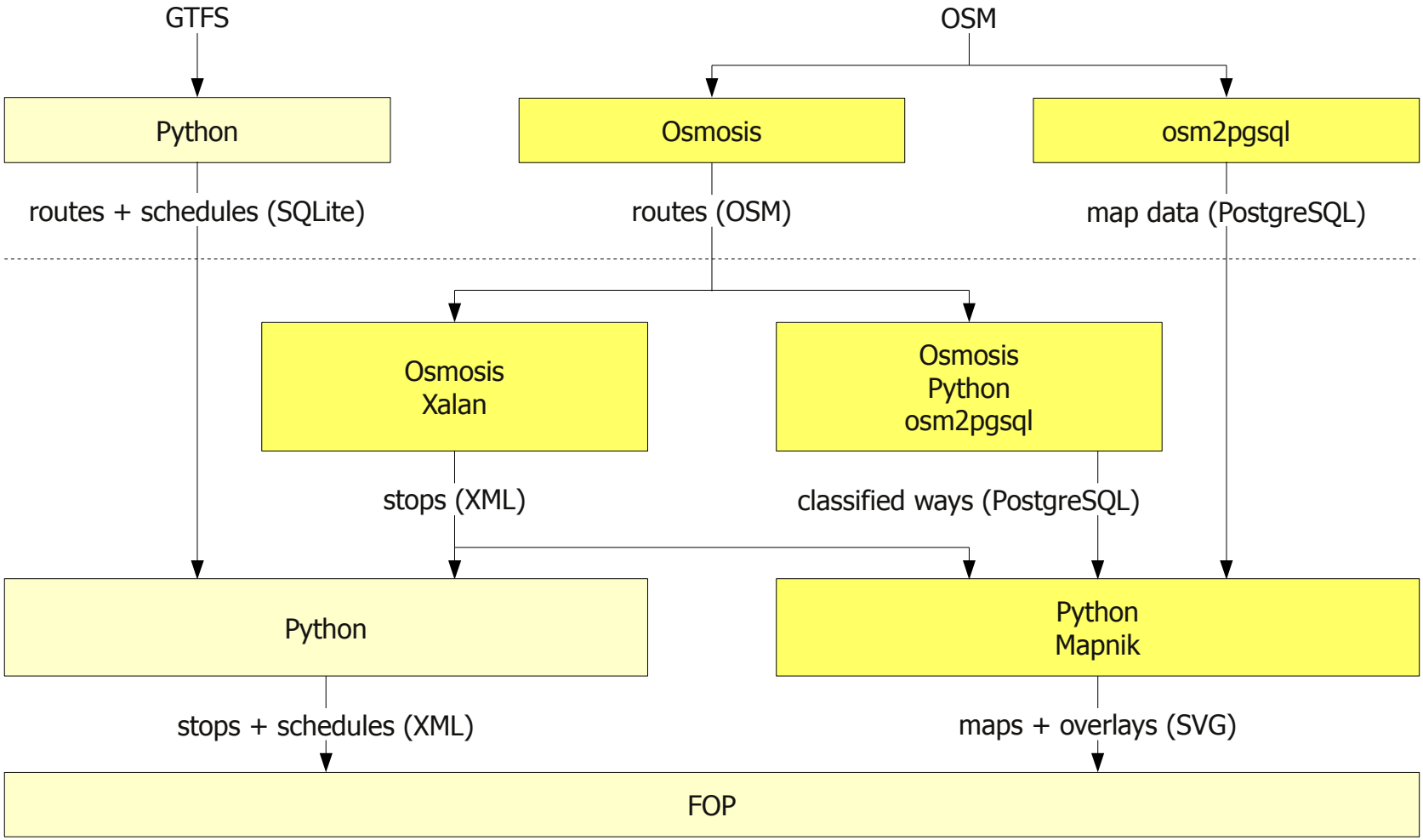

Layout by orngjce223, CC-BY

# Operator data



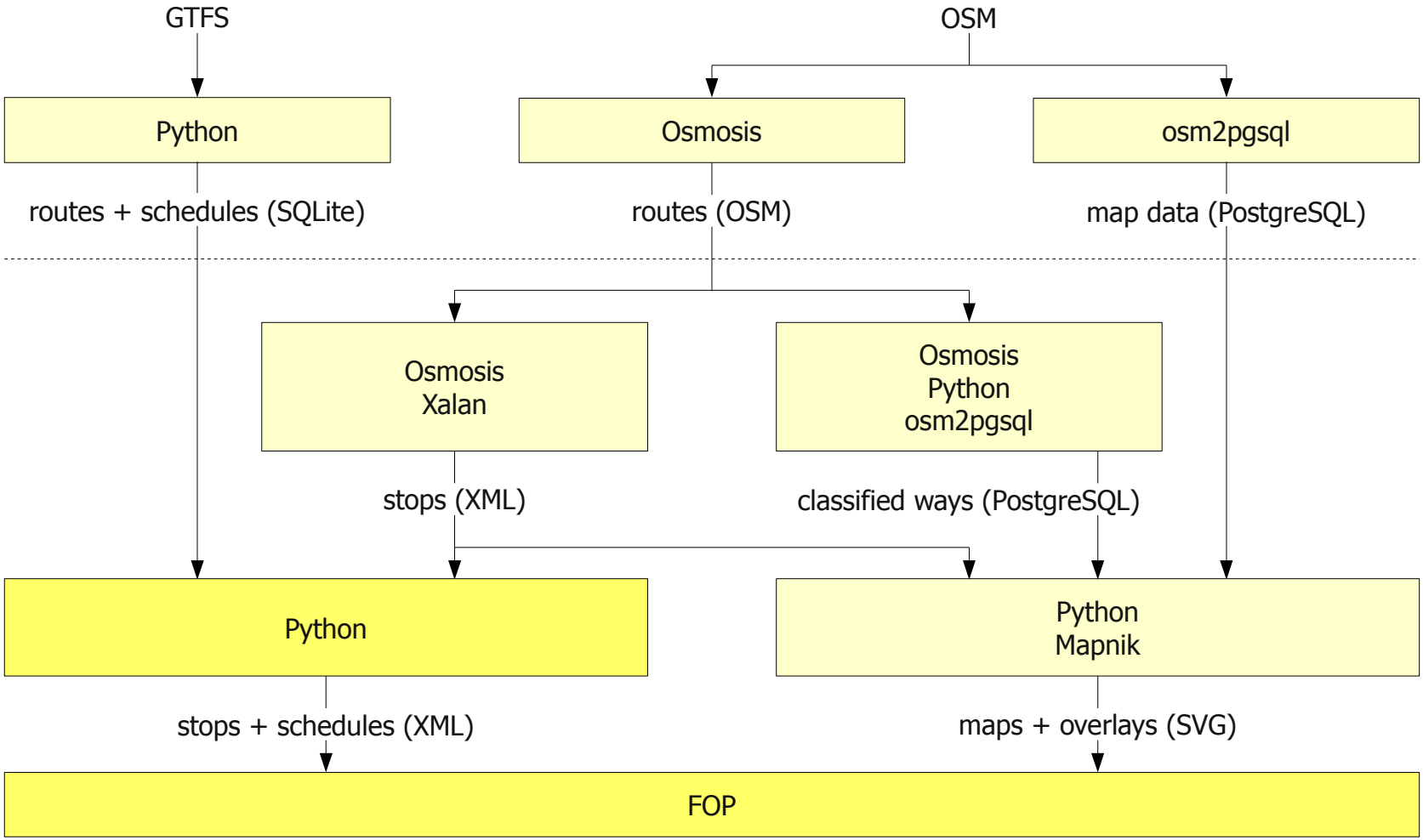

Layout by orngjce223, CC-BY

# OSM data



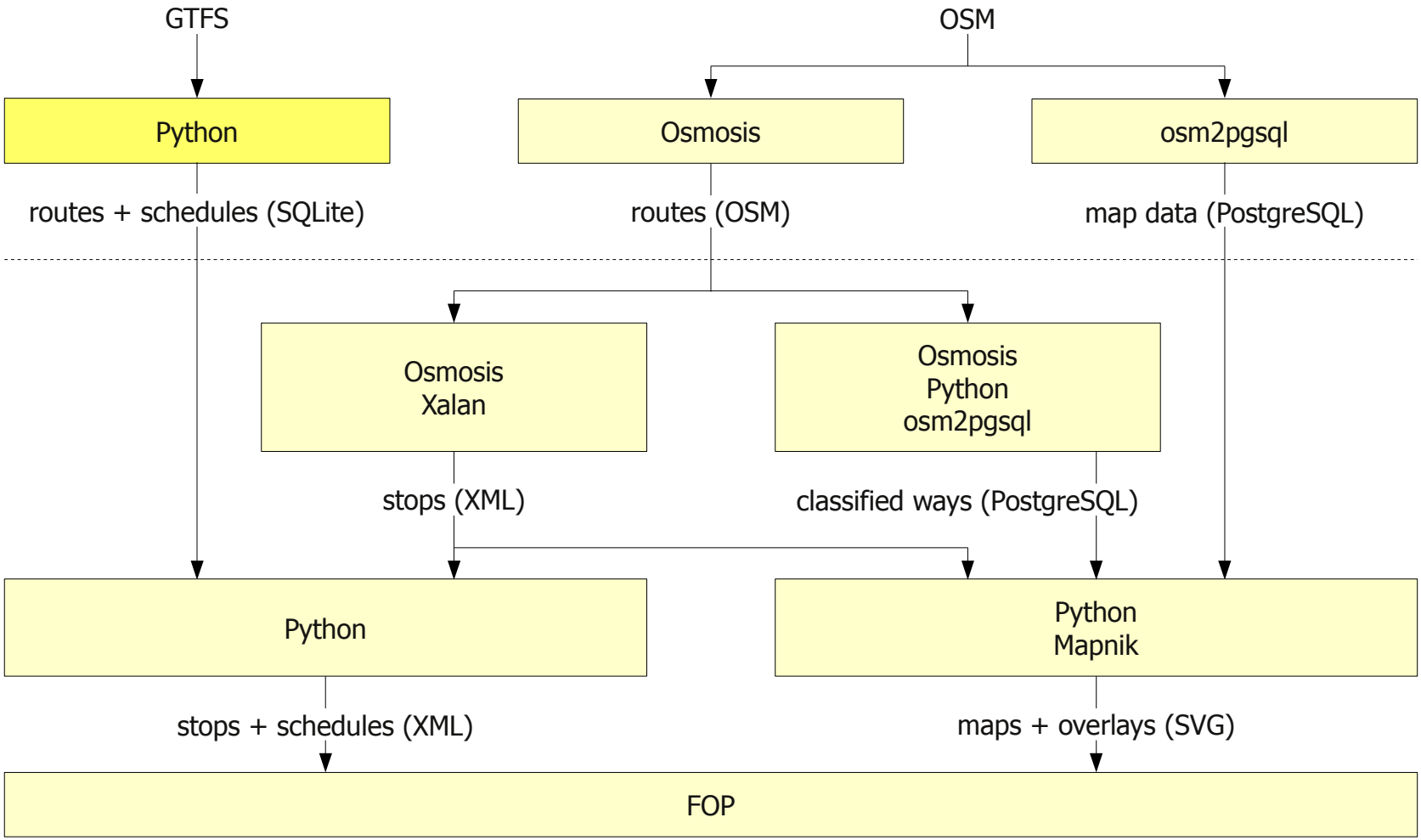

Layout by orngjce223, CC-BY

# Operator and OSM data




Layout by orngjce223, CC-BY

# GTFS preprocessing



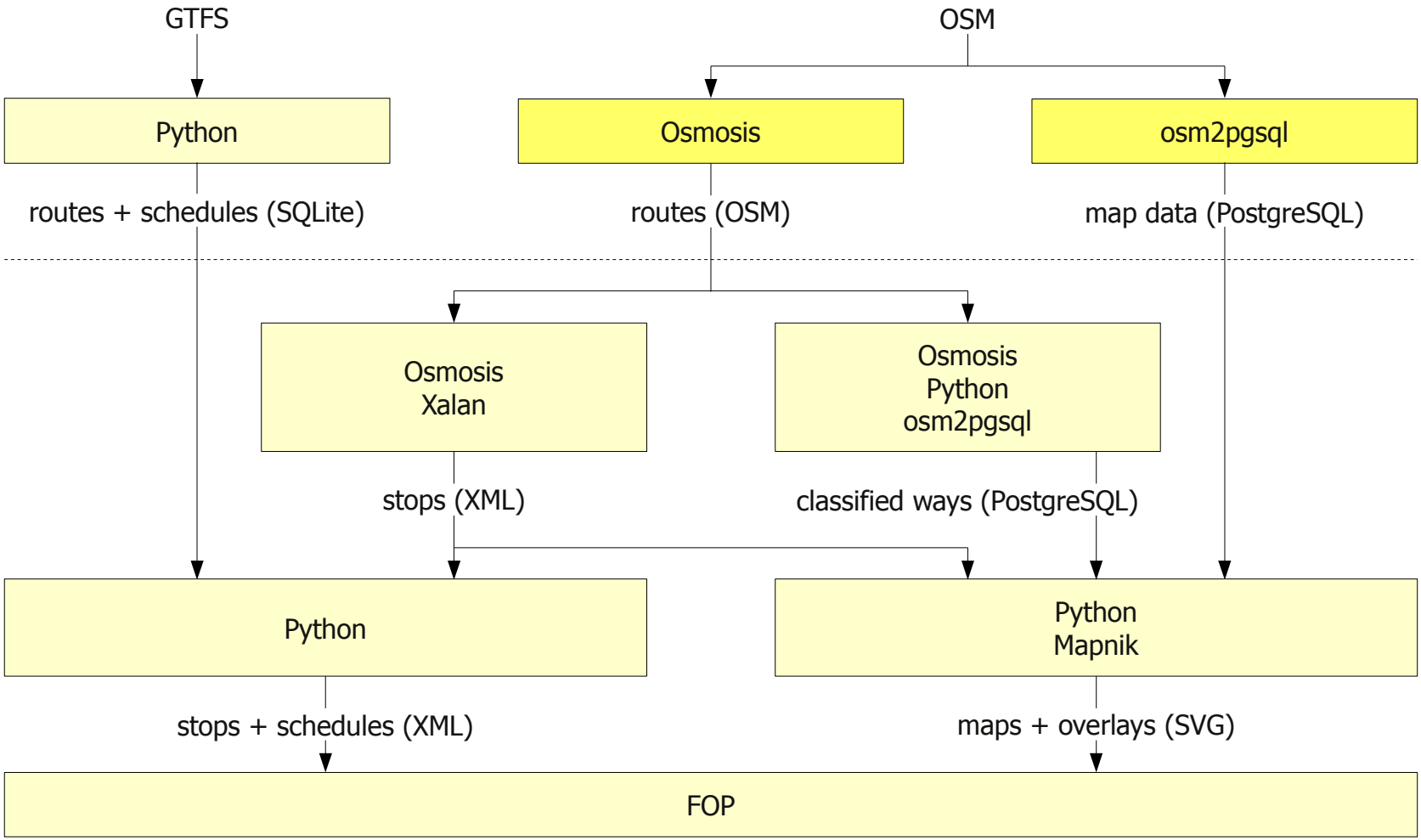

Layout by orngjce223, CC-BY

# GTFS preprocessing

- GTFS
  - Collection of CSV files
  - Route and stop information
- Information collection
  - Stops for each route
  - Schedules for each stop
  - Connecting routes for each stop

Layout by orngjce223, CC-BY

# OSM preprocessing



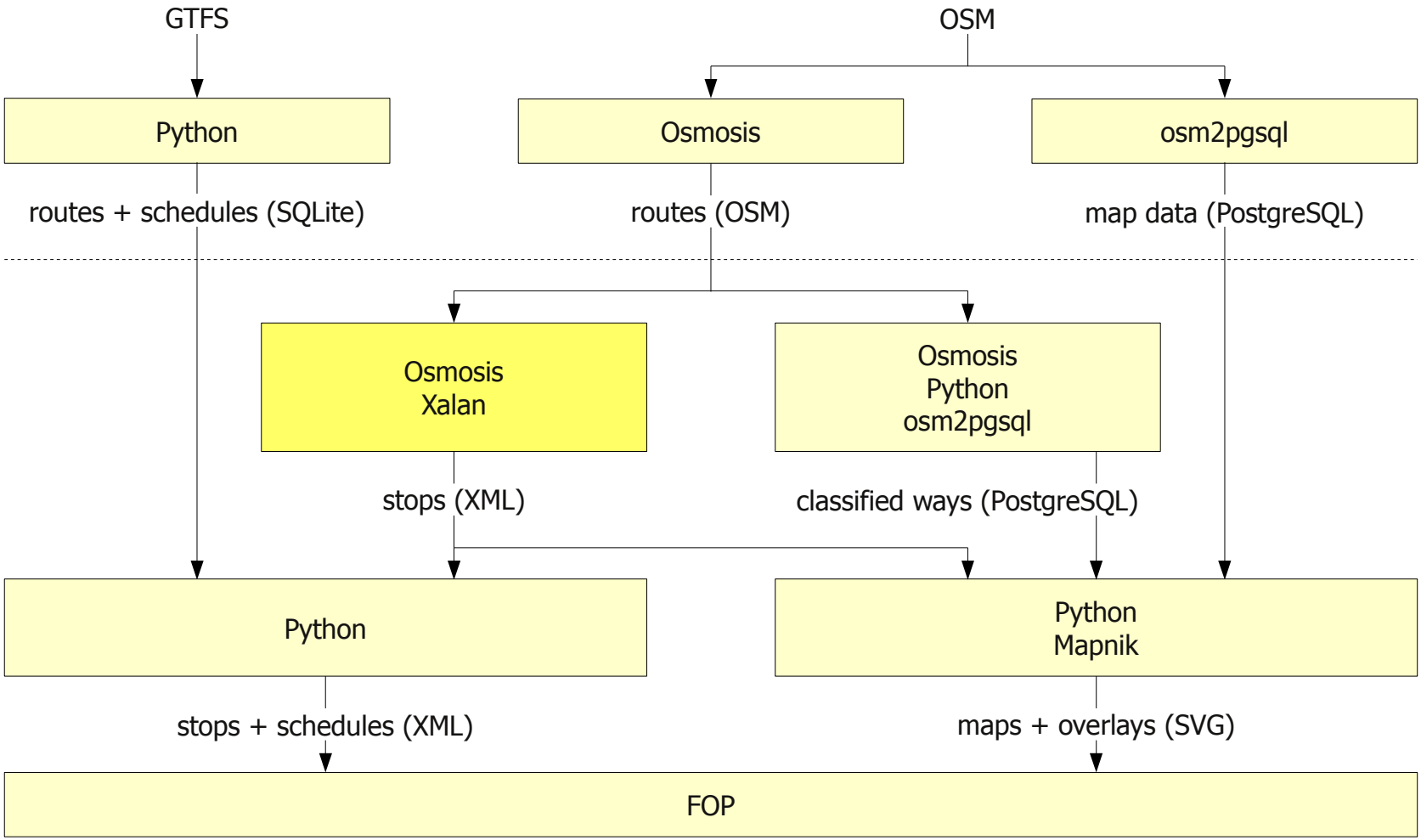

Layout by orngjce223, CC-BY

# OSM preprocessing

- Information collection
  - Route relations
  - Stop nodes
  - Using OSM public transport tags
  - Using OSM operator tag
- Import into database
  - Standard map data
  - Standard osm2pgsql
  - Standard database schema

Layout by orngjce223, CC-BY

# Stop list generation



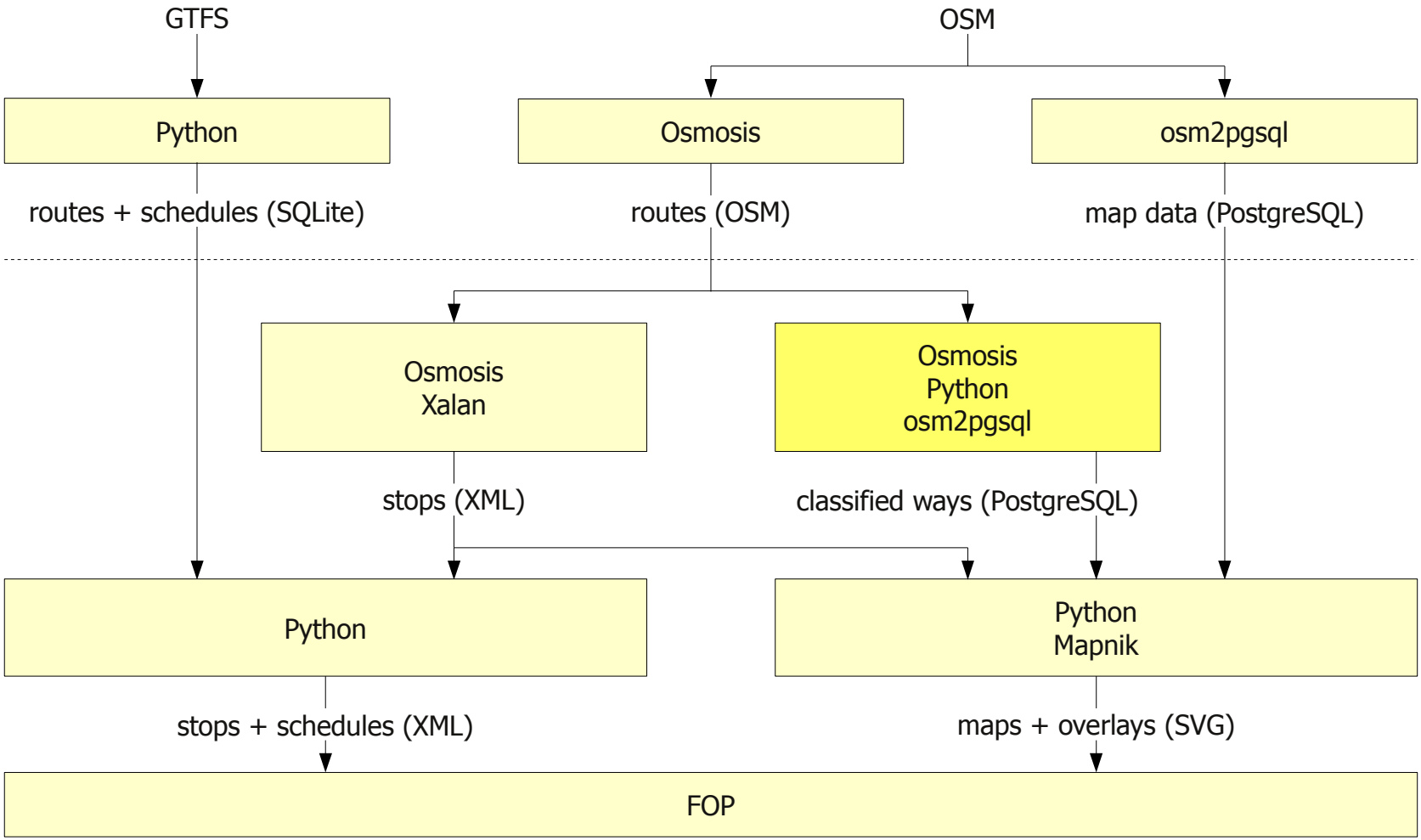

Layout by orngjce223, CC-BY

# Stop list generation

- Executed for specific route relation
- Information collection per stop
  - Location
  - Name
  - Reference
  - Attributes
    - Bench
    - Shelter
    - Tactile paving

Layout by orngjce223, CC-BY

# Way classification



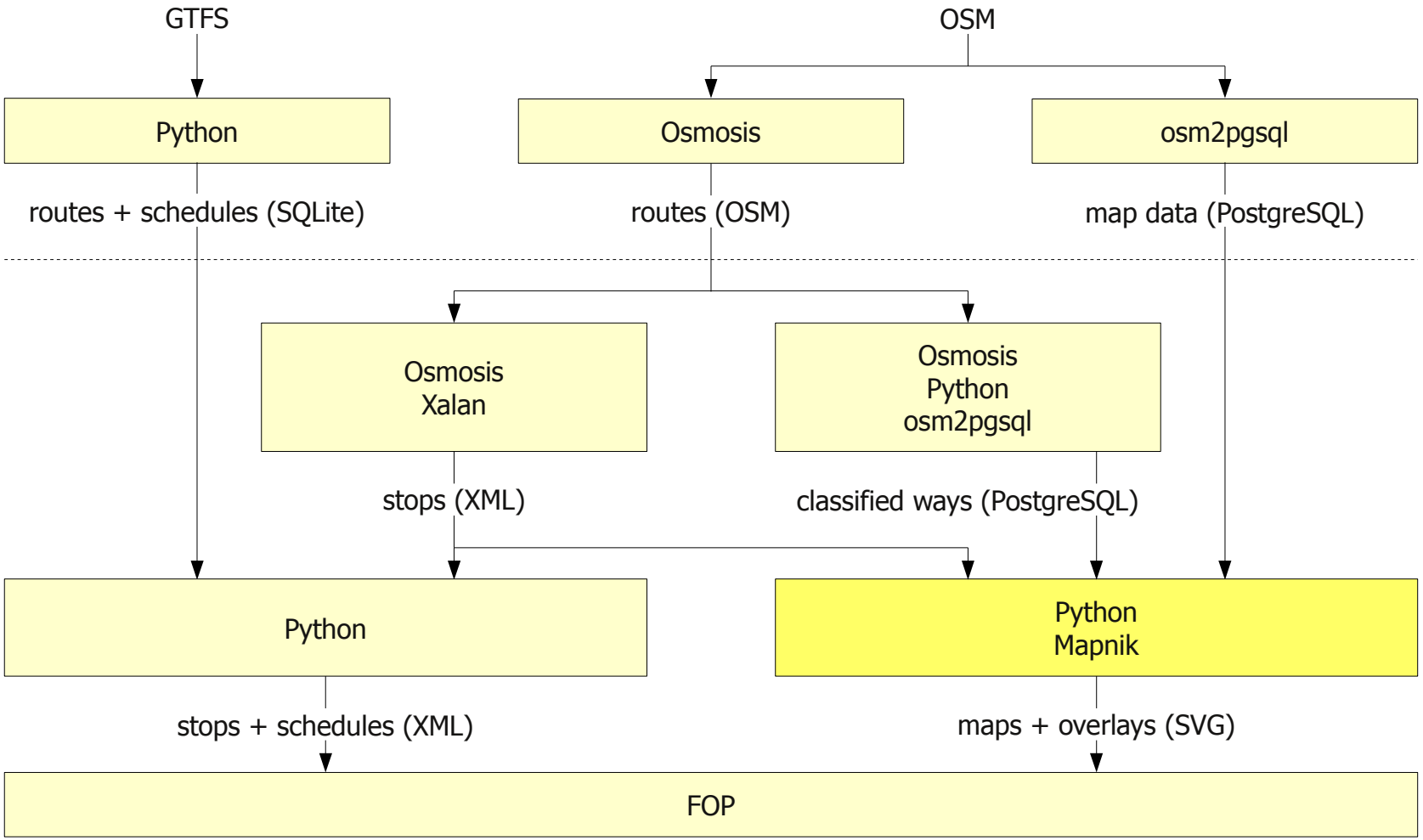

Layout by orngjce223, CC-BY

# Way classification

- Classification
  - Ways along current route relation
  - Ways along other route relations
- Import into database
  - Classified ways
  - Concatenated route references
  - Standard osm2pgsql
  - Standard database schema

Layout by orngjce223, CC-BY

# Map and overlay generation



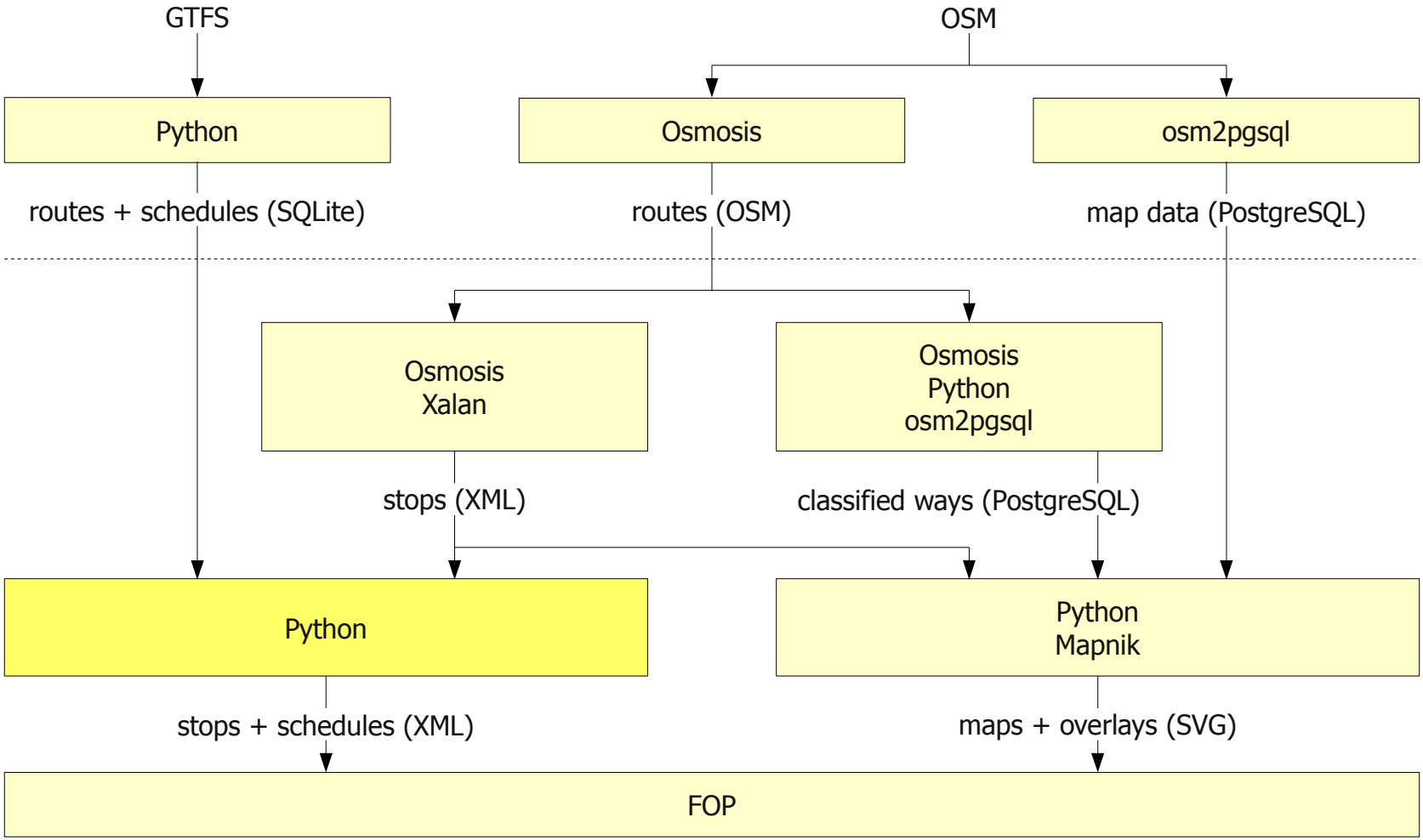

Layout by orngjce223, CC-BY

# Map and overlay generation

- Single SVG overview map
- SVG close-up map for each stop
  - Highlighting current route relation
  - Showing all other route relations
- SVG overview map overlay for each stop
  - Highlighting current stop
  - Showing all other stops

Layout by orngjce223, CC-BY

# Schedule addition



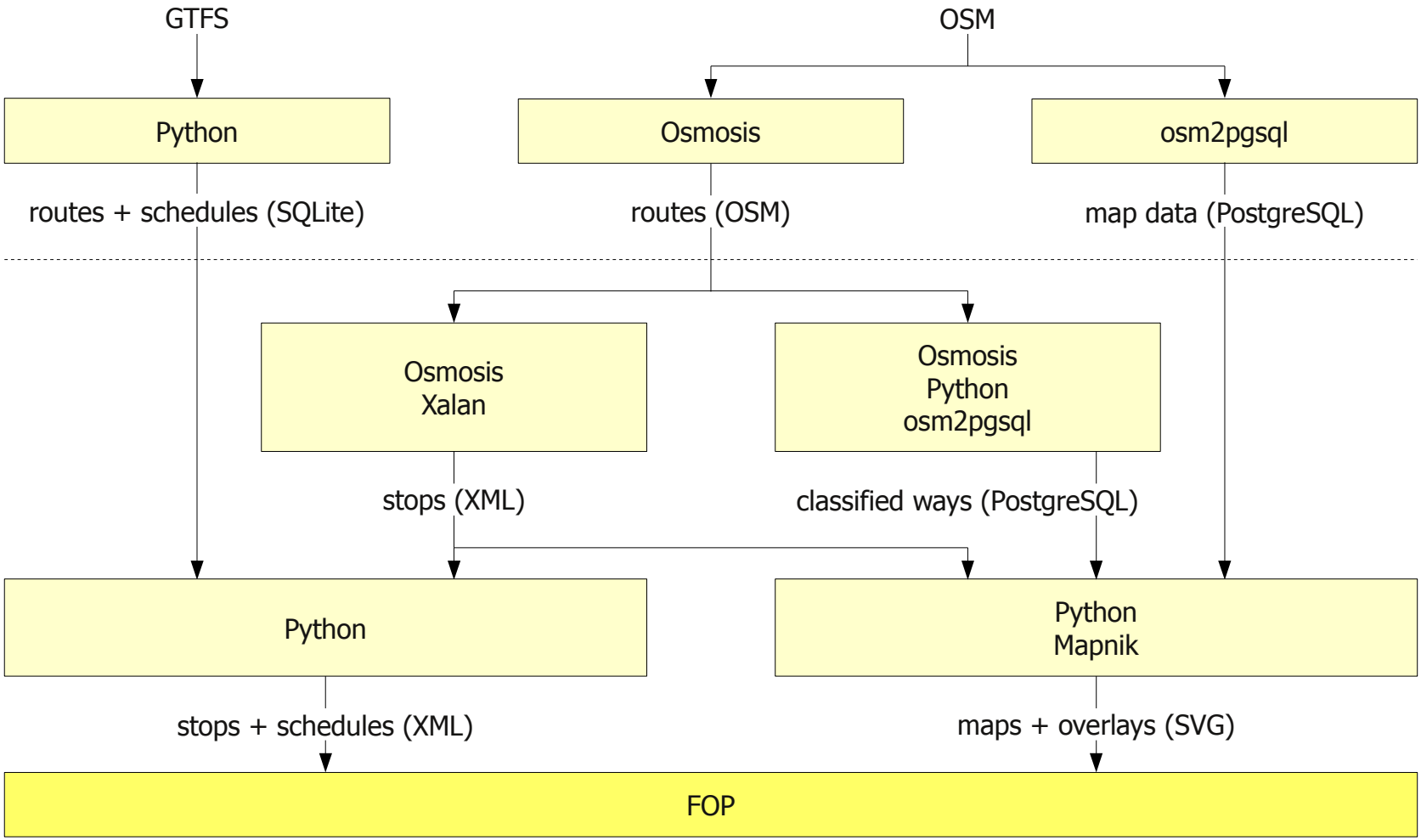

Layout by orngjce223, CC-BY

# Schedule addition

- Combination of operator and OSM data
  - Schedules from operator
  - Connections from operator
  - Stop names from OSM
  - Stop attributes from OSM
- Matching by route reference, stop reference

Layout by orngjce223, CC-BY

# Strip map and placard generation




Layout by orngjce223, CC-BY

# Strip map and placard generation

- One A4 page for each stop
- Generated using Apache FOP
- Strip map
  - Previous, current, subsequent stops
  - Stop names, references, attributes
  - Connections
- Overview map with overlay
- Close-up map
- Schedule

Layout by orngjce223, CC-BY

# Results

- Printed stop placards
  - Combined benefits of operator and OSM data
  - Automatic generation
  - Flexible presentation
- Challenges
  - OSM consistency
  - OSM completeness
  - Apache FOP

Layout by orngjce223, CC-BY

# Beyond public transport

- Custom printed documents
  - Driven by changes to external data
  - Incorporating changes to OSM data
- Benefits to external data provider
  - Maps
  - Flexible toolchain
- Benefits to OSM
  - Improved consistency and completeness
  - Project visibility

Layout by orngjce223, CC-BY

# Custom printed OSM maps: Public transport and beyond

Dr Bartosz Fabianowski
bartosz@dobini.com

Dobini
www.dobini.com

Layout by orngjce223, CC-BY